

(6)原方程可化为 $x^2+6x=0$, $\therefore x(x+6)=0$, 则 $x=0$ 或 $x+6=0$, 解得 $x_1=0, x_2=-6$.

6. D 7. D 8. A 9. 6 10. -2 -15 11. -3 或 4

12. $2(x+3)(x-2)=0$ (答案不唯一)

13. 解: (1) $3(x+4)(x+2)=2(x+2)$, 移项, 得 $(x+2)(3x+10)=0$, 所以 $x_1=-2, x_2=-\frac{10}{3}$.

(2) $x^2-2\sqrt{6}x+6=0$, 即 $(x-\sqrt{6})^2=0$, 所以 $x_1=x_2=\sqrt{6}$.

(3) $(3x-1+2x)(3x-1-2x)=0$, 整理, 得 $(5x-1)(x-1)=0$, 所以 $x_1=\frac{1}{5}, x_2=1$.

(4) $(1-y)^2-6(1-y)+9=0$, 所以 $(1-y-3)^2=0$, 所以 $y_1=y_2=-2$.

14. 解: $\because$ 方程 $(m-2)x^2-2(m-1)x+\frac{1}{2}m^2=0$ 有一个根为 1, $\therefore (m-2)-2(m-1)+\frac{1}{2}m^2=0$,

即 $\frac{1}{2}m^2-m=0$, 解得 $m=0$ 或 $m=2$.

当 $m=0$ 时, 方程为 $-2x^2+2x=0$, 它的根为 $x_1=0, x_2=1$;

当 $m=2$ 时, 方程为 $-2x+2=0$, 它的根为 $x=1$.

15. (1) 2 4

(2) 解: $x^2-3x-4=x^2+(-4+1)x+(-4)\times 1=(x+1)(x-4)=0$, 即 $x+1=0$ 或 $x-4=0$, 解得 $x_1=-1, x_2=4$.

(3) 解: 设 $2y-1=a$, 则 $a^2-8a+12=0$, 即 $a^2-8a+12=a^2+(-6-2)a+(-6)\times(-2)=(a-6)(a-2)=0$,

解得 $a=6$ 或 $a=2$, 即 $2y-1=6$ 或 $2y-1=2$,

解得 $y_1=\frac{7}{2}, y_2=\frac{3}{2}$.

专题(1) 一元二次方程的解法

1. 解: (1) $\because (x-3)^2-4=0, \therefore (x-3)^2=4, \therefore x-3=\pm 2$, $\therefore x_1=1, x_2=5$.

(2) $\because (2x-1)^2=25, \therefore 2x-1=\pm 5$, $\therefore x_1=3, x_2=-2$.

(3) $\because 3(x+2)^2=\frac{1}{3}, \therefore (x+2)^2=\frac{1}{9}, \therefore x+2=\pm \frac{1}{3}$, $\therefore x_1=-\frac{5}{3}, x_2=-\frac{7}{3}$.

(4) $\because (3x-1)^2=(x+1)^2, \therefore 3x-1=\pm(x+1)$, $\therefore 2x=2$ 或 $4x=0, \therefore x_1=1, x_2=0$.

2. 解: (1) 移项, 得 $x^2-4x=1$, 配方, 得 $x^2-4x+4=1+4$,

即 $(x-2)^2=5$, 所以 $x-2=\pm\sqrt{5}$, 所以 $x_1=2+\sqrt{5}, x_2=2-\sqrt{5}$.

(2) 移项, 得 $x^2+3x=4$,

配方, 得 $x^2+3x+(\frac{3}{2})^2=4+(\frac{3}{2})^2$,

即 $(x+\frac{3}{2})^2=\frac{25}{4}$,

所以 $x+\frac{3}{2}=\pm\frac{5}{2}$, 所以 $x_1=1, x_2=-4$.

(3) 化二次项系数为 1, 得 $x^2-2x-4=0$,

移项, 得 $x^2-2x=4$,

配方, 得 $x^2-2x+1=4+1$,

即 $(x-1)^2=5$, 所以 $x-1=\pm\sqrt{5}$,

所以 $x_1=1+\sqrt{5}, x_2=1-\sqrt{5}$.

(4) 化二次项系数为 1, 得 $x^2-\frac{5}{2}x+1=0$, 移项, 得 $x^2-\frac{5}{2}x=-1$, 配方, 得 $x^2-\frac{5}{2}x+(\frac{5}{4})^2=-1+(\frac{5}{4})^2$,

即 $(x-\frac{5}{4})^2=\frac{9}{16}$, 所以 $x-\frac{5}{4}=\pm\frac{3}{4}$, 所以 $x_1=2, x_2=\frac{1}{2}$.

3. 解: (1) $\because a=1, b=-5, c=1$, $\therefore b^2-4ac=(-5)^2-4\times 1\times 1=21>0$,

$\therefore x=\frac{-(-5)\pm\sqrt{21}}{2\times 1}, \therefore x_1=\frac{5+\sqrt{21}}{2}, x_2=\frac{5-\sqrt{21}}{2}$.

(2) $\because a=1, b=-2\sqrt{2}, c=2$,

$\therefore b^2-4ac=(-2\sqrt{2})^2-4\times 1\times 2=0$,

$\therefore x=\frac{-(-2\sqrt{2})\pm 0}{2\times 1}=\sqrt{2}, \therefore x_1=x_2=\sqrt{2}$.

(3) 化方程为一般形式, 得 $x^2+3x+4=0$.

$\because a=1, b=3, c=4$,

$\therefore b^2-4ac=3^2-4\times 1\times 4=9-16=-7<0$, $\therefore$ 此方程没有实数根.

(4) $\because a=1, b=m, c=-2m^2$,

$\therefore b^2-4ac=m^2-4\times 1\times (-2m^2)=9m^2$,

$\therefore x=\frac{-m\pm 3m}{2\times 1}, \therefore x_1=-2m, x_2=m$.

$\therefore$ 此方程没有实数根.

(4) $\because a=1, b=m, c=-2m^2$,

$\therefore b^2-4ac=m^2-4\times 1\times (-2m^2)=9m^2$,

$\therefore x=\frac{-m\pm 3m}{2\times 1}, \therefore x_1=-2m, x_2=m$.

4. 解: (1) 原方程可化为 $x(5x-4)=0$, 所以 $x=0$ 或 $5x-4=0$, 所以 $x_1=0, x_2=\frac{4}{5}$.

(2) 移项, 得 $x(x-6)+4(x-6)=0$, 即 $(x-6)(x+4)=0$, 所以 $x-6=0$ 或 $x+4=0$, 所以 $x_1=6, x_2=-4$.

(3) 移项, 得 $x(x-3)-(x-3)=0$, 因式分解, 得 $(x-3)(x-1)=0$, 则 $x-3=0$ 或 $x-1=0$, 所以 $x_1=3, x_2=1$.

(4) 因式分解, 得 $[2(2x+1)+3(2x-1)][2(2x+1)-$

$3(2x-1)=0$, 即 $(4x+2+6x-3)(4x+2-6x+3)=0$,
 $(10x-1)(-2x+5)=0$, 则 $10x-1=0$ 或 $-2x+5=0$, 所以
 $x_1=\frac{1}{10}, x_2=\frac{5}{2}$.

5. 解: (1) $(x+2+3)(x+2-3)=0$,

$\therefore x+5=0$ 或 $x-1=0$, $\therefore x_1=-5, x_2=1$.

(2) 原方程移项, 得 $x^2-2x=4$, 配方, 得 $x^2-2x+1=5$,

即 $(x-1)^2=5$, $\therefore x=1\pm\sqrt{5}$, $\therefore x_1=1+\sqrt{5}, x_2=1-\sqrt{5}$.

(3) $\because a=1, b=3, c=1, \therefore b^2-4ac=3^2-4\times 1\times 1=5$,

$\therefore x=\frac{-3\pm\sqrt{5}}{2}, \therefore x_1=\frac{-3+\sqrt{5}}{2}, x_2=\frac{-3-\sqrt{5}}{2}$.

(4) 移项, 得 $4x^2-8x=-1$,

方程两边都除以 4, 得 $x^2-2x=-\frac{1}{4}$,

配方, 得 $x^2-2x+1^2=-\frac{1}{4}+1^2$, 即 $(x-1)^2=\frac{3}{4}$,

开方, 得 $x-1=\pm\frac{\sqrt{3}}{2}, \therefore x_1=\frac{2+\sqrt{3}}{2}, x_2=\frac{2-\sqrt{3}}{2}$.

(5) 原方程整理, 得 $x^2+4x-12=0$,

所以 $(x+6)(x-2)=0$, 则 $x+6=0$ 或 $x-2=0$, 所以
 $x_1=-6, x_2=2$.

(6) 设 $y-3=t$, 则原方程可化为 $t^2+3t+2=0$, 解得 $t_1=-1$,
 $t_2=-2$, 所以 $y-3=-1$ 或 $y-3=-2$, 所以 $y_1=2$,
 $y_2=1$.

6. 解: (1) $\triangle ABC$ 为直角三角形, 理由如下:

$\because$ 方程有两个相等的实数根, $\therefore (-4b)^2-4(a+c)[-4(a-c)]=0$, $\therefore b^2+a^2=c^2$, $\therefore \triangle ABC$ 为直角三角形.

(2) $\because \triangle ABC$ 是等边三角形, $\therefore a=b=c$, $\therefore$ 原方程可化为
 $2ax^2-4ax=0$, $\because a\neq 0, \therefore x^2-2x=0, \therefore x_1=0, x_2=2$.

7. 解: 设 $y=x^2+x$, 则原方程可变为 $y^2-4y-12=0$,

解得 $y=6$ 或 $y=-2$.

当 $y=6$ 时, $x^2+x=6$, 解得 $x_1=-3, x_2=2$;

当 $y=-2$ 时, $x^2+x=-2$, 该方程无实数根.

综上所述, 原方程的解为 $x_1=-3, x_2=2$.

作业 8 一元二次方程的根与系数的关系

1. A 2. C 3. $\frac{1}{2}$ 4. 4

5. 解: (1) 根据题意, 得 $36+4m\geq 0$, 解得 $m\geq -9$, 即 m 的取值范围为 $m\geq -9$.

(2) 根据题意, 得 $x_1+x_2=-6, x_1x_2=-m$,

$\because x_1+x_2-2x_1x_2=0, \therefore -6-2\times(-m)=0$, 解得 $m=3$,
 即 m 的值为 3.

6. C 7. C 8. B 9. $x^2-4x+2=0$ (答案不唯一)

10. 5

11. 解: (1) $\because$ 方程有实数根, $\therefore b^2-4ac=25-4m\geq 0$, 解得

$$m\leq \frac{25}{4}.$$

(2) 由一元二次方程根与系数的关系可知 $x_1+x_2=5$,

$x_1x_2=m, \because 3x_1-2x_2=5, \therefore 3x_1+3x_2-5x_2=5$,

$\therefore -5x_2=-10$, 解得 $x_2=2$. 把 $x=2$ 代入原方程, 得

$2^2-5\times 2+m=0$, 解得 $m=6$.

12. (1) 证明: $b^2-4ac=[-(m+2)]^2-4\times 1\times(m-1)=$
 $m^2+4m+4-4m+4=m^2+8$.

$\because m^2\geq 0, \therefore b^2-4ac>0$,

$\therefore$ 无论 m 取何值, 方程都有两个不相等的实数根.

(2) 解: 方程 $x^2-(m+2)x+m-1=0$ 的两个实数根为

x_1, x_2 ,

则 $x_1+x_2=m+2, x_1x_2=m-1$.

$\because x_1^2+x_2^2-x_1x_2=9$, 即 $(x_1+x_2)^2-3x_1x_2=9$,

$\therefore (m+2)^2-3(m-1)=9$,

整理, 得 $m^2+m-2=0$,

$\therefore (m+2)(m-1)=0$,

解得 $m_1=-2, m_2=1$,

$\therefore m$ 的值为 -2 或 1 .

13. (1) $-\frac{3}{2} -\frac{1}{2}$

(2) 解: $\because$ 一元二次方程 $2x^2+3x-1=0$ 的两根分别为

$m, n, \therefore m+n=-\frac{3}{2}, mn=-\frac{1}{2}$,

$\therefore m^2+n^2=(m+n)^2-2mn=\frac{9}{4}+1=\frac{13}{4}$.

(3) 解: $\because$ 实数 s, t 满足 $2s^2+3s-1=0, 2t^2+3t-1=0$,
 且 $s\neq t, \therefore s, t$ 是一元二次方程 $2x^2+3x-1=0$ 的两个

实数根, $\therefore s+t=-\frac{3}{2}, st=-\frac{1}{2}. \because (t-s)^2=(t+s)^2-$

$4st=(-\frac{3}{2})^2-4\times(-\frac{1}{2})=\frac{17}{4}, \therefore t-s=\pm\frac{\sqrt{17}}{2}$,

$\therefore \frac{1}{s}-\frac{1}{t}=\frac{t-s}{st}=\frac{\pm\frac{\sqrt{17}}{2}}{-\frac{1}{2}}=\pm\sqrt{17}$.

周练(1)

1. A 2. D 3. B 4. D 5. A 6. 1 (答案不唯一)

7. 2 (答案不唯一, $t>1$ 即可) 8. 2 9. 3 或 4 10. 6

11. 解: (1) $(x+1)^2=3, x+1=\pm\sqrt{3}$, 解得 $x_1=\sqrt{3}-1, x_2=-\sqrt{3}-1$.

(2) $x^2-3x=x-2, x^2-4x=-2, x^2-4x+4=2$, 即

$(x-2)^2=2, x-2=\sqrt{2}$ 或 $x-2=-\sqrt{2}$, 解得 $x_1=2+\sqrt{2}, x_2=2-\sqrt{2}$.

(3) $3x-1=\pm(x-1)$, 即 $3x-1=x-1$ 或 $3x-1=$

4. 解: (1) 设每千克核桃应降价 x 元,

根据题意, 得 $(60-x-40)(100+\frac{x}{3} \times 30) = 2090$,

化简, 得 $x^2 - 10x + 9 = 0$, 解得 $x_1 = 1, x_2 = 9$.

答: 每千克核桃应降价 1 元或 9 元.

(2) 由 (1) 可知每千克核桃可降价 1 元或 9 元.

因为要尽可能让利于顾客, 所以每千克核桃应降价 9 元,
 $60 - 9 = 51$ (元),

则 $\frac{51}{60} \times 100\% = 85\%$.

答: 该店应按原售价的八五折出售.

5. B 6. 50 7. 25

8. 解: 设该班参加这次研学活动的学生有 x 人,

$\because 30 \times 80 = 2400 < 2800, \therefore x > 30$.

根据题意, 得 $x[80 - (x - 30)] = 2800$,

整理, 得 $x^2 - 110x + 2800 = 0$,

解得 $x_1 = 40, x_2 = 70$.

当 $x = 70$ 时, $80 - (x - 30) = 40, 40 < 60$, 不符合题意, 舍去.

答: 该班参加这次研学活动的学生有 40 人.

9. (1) $(60 - \frac{x}{10})$ (2) $(-\frac{1}{10}x^2 + 40x + 12000)$

(3) 解: 根据题意, 得 $(200 + x)(60 - \frac{x}{10}) - 20 \times (60 - \frac{x}{10}) = 14000$, 整理, 得 $x^2 - 420x + 32000 = 0$, 解得 $x = 320$ 或 $x = 100$.

当 $x = 320$ 时, $200 + x = 520$ (不合题意, 舍去);

当 $x = 100$ 时, $200 + x = 300$.

答: 每个房间的定价应为 300 元.

10. (1) $(14 - \frac{x}{2})$

(2) 解: 设甲每个的售价提高 x 元时, 销售这两种商品当天的总利润是 268 元, 根据题意, 得

$(5+x)(30-2x) + (6+2x)(14-\frac{x}{2}-7) = 268$,

整理, 得 $3x^2 - 31x + 76 = 0$, 解得 $x_1 = 4, x_2 = \frac{19}{3}$.

$\because$ 售价均为整数, $\therefore x = 4$.

答: 当甲每个的售价提高 4 元时, 销售这两种商品当天的总利润是 268 元.

作业 11 用一元二次方程解决问题 (3)

1. B 2. 6 或 8

3. 解: 设它们爬行的时间为 t s, 则 $BM = (12 - 2t)$ cm, $BN = (8 - t)$ cm. 根据题意, 得 $\frac{1}{2}(12 - 2t)(8 - t) = 24$, 整理, 得

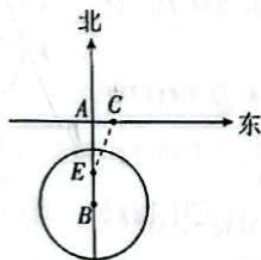
$t^2 - 14t + 24 = 0$, 解得 $t_1 = 2, t_2 = 12$. 当 $t = 12$ 时, $12 - 2t = -12 < 0$, $\therefore t = 12$ 不合题意, 舍去.

答: 它们爬行的时间为 2 s.

4. A 5. 0 或 0.4

6. 解: 如答图, 设轮船最初遇到台风的时间为 t h, 此时轮船位于 C 处, 台风中心移到 E 处, 连接 CE , 则 $AC = 20t$ 海里, $AE = AB - BE = (100 - 40t)$ 海里, $AC^2 + AE^2 = EC^2$, $\therefore (20t)^2 + (100 - 40t)^2 = (20\sqrt{10})^2$, 化简, 得 $t^2 - 4t + 3 = 0$, 解得 $t_1 = 1, t_2 = 3$ (不合题意, 舍去).

答: 轮船最初遇到台风的时间为 1 h.



第 6 题答图

7. 解: (1) 根据题意, 得 $\frac{1}{2}t^2 + \frac{3}{2}t + 4t = 21$,

解得 $t = 3$ 或 $t = -14$ (不合题意, 舍去).

答: 甲、乙从开始运动到第一次相遇时, 运动了 3 s.

(2) 根据题意, 得 $\frac{1}{2}t^2 + \frac{3}{2}t + 4t = 3 \times 21$, 解得 $t = 7$ 或 $t = -18$ (不合题意, 舍去).

答: 甲、乙从开始运动到第二次相遇时, 运动了 7 s.

8. 解: (1) 当点 P 在 BC 上时, $0 \leq t \leq 2$, $AQ = t$ cm, $BQ = (4 - t)$ cm, $BP = 2t$ cm, $PC = (4 - 2t)$ cm.

$\because S_{\triangle PDQ} = S_{\text{正方形 } ABCD} - S_{\triangle ADQ} - S_{\triangle BPQ} - S_{\triangle CPD} = 11$,

$\therefore 16 - \frac{1}{2} \cdot 4 \cdot t - \frac{1}{2} \cdot (4 - t) \cdot 2t - \frac{1}{2} \cdot 4 \cdot (4 - 2t) = 11$,

整理, 得 $t^2 - 2t - 3 = 0$, 解得 $t_1 = -1, t_2 = 3$, 均不合题意, 舍去;

当点 P 在 CD 上时, $2 < t \leq 4$, $DP = (8 - 2t)$ cm,

$\because S_{\triangle PDQ} = \frac{1}{2}BC \cdot DP$, $\therefore \frac{1}{2} \times 4(8 - 2t) = 11$,

解得 $t = \frac{5}{4}$ (不合题意, 舍去).

$\therefore$ 不存在 t 的值, 使 $\triangle PQD$ 的面积为 11 cm^2 .

(2) 存在. 由题意, 知 $AQ = t$ cm, $BQ = (4 - t)$ cm, $BP = 2t$ cm, $PC = (4 - 2t)$ cm.

当 $DP = DQ$ 时, $\because DC = DA$, $\therefore \text{Rt} \triangle DPC \cong \text{Rt} \triangle DQA$,

$\therefore PC = AQ$, 即 $4 - 2t = t$, 解得 $t = \frac{4}{3}$;

当 $PD = PQ$ 时, 在 $\text{Rt} \triangle PBQ$ 中, $PQ^2 = PB^2 + BQ^2 = (2t)^2 + (4 - t)^2$,

在 $\text{Rt} \triangle PCD$ 中, $PD^2 = PC^2 + CD^2 = (4 - 2t)^2 + 4^2$,

$$\therefore (2t)^2 + (4-t)^2 = (4-2t)^2 + 4^2,$$

整理,得 $t^2 + 8t - 16 = 0$, 解得 $t_1 = -4\sqrt{2} - 4$ (舍去), $t_2 = 4\sqrt{2} - 4$.

综上,当 t 的值为 $\frac{4}{3}$ 或 $4\sqrt{2} - 4$ 时, $\triangle PQD$ 是以 PD 为腰的等腰三角形.

专题(2) 一元二次方程的应用

1. 1

2. 解: 如答图, 延长 EF 交 DA 的延长线于点 Q , 连接 DE , 设 $BE = x$.

$\because$ 四边形 $ABCD$ 是平行四边形, $\therefore DQ \parallel BC$,

$$\therefore \angle Q = \angle BEF.$$

$$\because AF = FB, \angle AFQ = \angle BFE,$$

$$\therefore \triangle QFA \cong \triangle EFB (\text{AAS}), \therefore AQ = BE = x, FQ = FE.$$

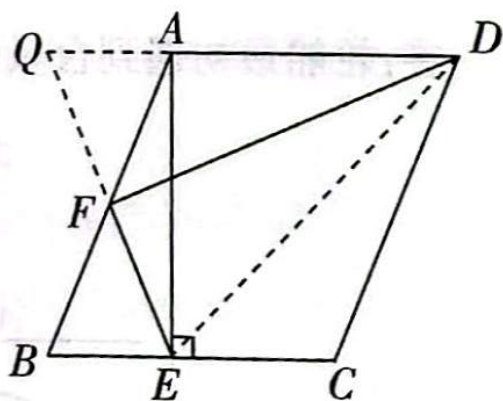
$$\because \angle EFD = 90^\circ, \therefore DF \perp QE, \therefore DQ = DE = x + 2.$$

$$\because AE \perp BC, BC \parallel AD, \therefore AE \perp AD,$$

$$\therefore \angle AEB = \angle EAD = 90^\circ.$$

$$\because AE^2 = DE^2 - AD^2 = AB^2 - BE^2, \therefore (x+2)^2 - 4 = 6 - x^2, \text{整理,得 } x^2 + 2x - 3 = 0, \text{解得 } x = 1 \text{ 或 } x = -3 (\text{舍去}),$$

$$\therefore BE = 1, \therefore AE = \sqrt{AB^2 - BE^2} = \sqrt{(\sqrt{6})^2 - 1^2} = \sqrt{5}.$$



第2题答图

3. 20%

$$\therefore m^2 + n^2 = (m+n)^2 - 2mn = \frac{9}{4} + 1 = \frac{13}{4}.$$

(3)解: $\because$ 实数 s, t 满足 $2s^2 + 3s - 1 = 0, 2t^2 + 3t - 1 = 0$,
且 $s \neq t, \therefore s, t$ 是一元二次方程 $2x^2 + 3x - 1 = 0$ 的两个
实数根, $\therefore s + t = -\frac{3}{2}, st = -\frac{1}{2}. \because (t-s)^2 = (t+s)^2 -$
 $4st = \left(-\frac{3}{2}\right)^2 - 4 \times \left(-\frac{1}{2}\right) = \frac{17}{4}, \therefore t - s = \pm \frac{\sqrt{17}}{2},$

$$\therefore \frac{1}{s} - \frac{1}{t} = \frac{t-s}{st} = \frac{\pm \frac{\sqrt{17}}{2}}{-\frac{1}{2}} = \pm \sqrt{17}.$$

周练(1)

1. A 2. D 3. B 4. D 5. A 6. 1(答案不唯一)

7. 2(答案不唯一, $t > 1$ 即可) 8. 2 9. 3 或 4 10. 6

11. 解: (1) $(x+1)^2 = 3, x+1 = \pm\sqrt{3}$, 解得 $x_1 = \sqrt{3} - 1, x_2 =$
 $-\sqrt{3} - 1.$

(2) $x^2 - 3x = x - 2, x^2 - 4x = -2, x^2 - 4x + 4 = 2$, 即
 $(x-2)^2 = 2, x-2 = \sqrt{2}$ 或 $x-2 = -\sqrt{2}$, 解得 $x_1 = 2 +$
 $\sqrt{2}, x_2 = 2 - \sqrt{2}.$

(3) $3x - 1 = \pm(x - 1)$, 即 $3x - 1 = x - 1$ 或 $3x - 1 =$

参考答案 5

$$-(x-1), \text{解得 } x_1=0, x_2=\frac{1}{2}.$$

$$(4) x^2-4x=2, x^2-4x+4=6, \text{即 } (x-2)^2=6, x-2=\pm\sqrt{6}, \text{解得 } x_1=2+\sqrt{6}, x_2=2-\sqrt{6}.$$

$$(5) \text{方程整理, 得 } x^2+2x=4, x^2+2x+1=5, \text{即 } (x+1)^2=5, x+1=\pm\sqrt{5}, \text{解得 } x_1=-1+\sqrt{5}, x_2=-1-\sqrt{5}.$$

$$(6) \because a=2, b=-7, c=6, \therefore b^2-4ac=49-48=1, \therefore x=\frac{7\pm1}{4}, \text{解得 } x_1=2, x_2=\frac{3}{2}.$$

12. 解: (1) $\because$ 关于 x 的方程 $x^2-2mx+m^2-n=0$ 有两个不相等的实数根,

$$\therefore (-2m)^2-4(m^2-n)=4m^2-4m^2+4n>0, \therefore n>0.$$

(2) $\because n$ 为符合条件的最小整数, $n>0, \therefore n=1,$

$$\therefore \text{原方程为 } x^2-2mx+m^2-1=0.$$

设该方程的根是 $a, 2a,$

$$\therefore a+2a=2m, a \cdot 2a=m^2-1,$$

解得 $a=2, m=3$ 或 $a=-2, m=-3$ (不合题意, 舍去),

$\therefore m$ 的值为 3.

13. 解: (1) $\because \square ABCD$ 的两边 AB, AD 的长是关于 x 的方程 $x^2-ax+a-1=0$ 的两个实数根, 且 AB 的长为 2,

$\therefore x=2$ 是关于 x 的方程 $x^2-ax+a-1=0$ 的实数根.

将 $x=2$ 代入, 得 $2^2-2a+a-1=0$, 解得 $a=3,$

$\therefore$ 原方程为 $x^2-3x+2=0$, 即 $(x-1)(x-2)=0,$

解得 $x_1=2, x_2=1, \therefore AD$ 的长为 1.

(2) $\because$ 菱形 $ABCD$ 的两边 AB, AD 的长是关于 x 的方程 $x^2-ax+a-1=0$ 的两个实数根,

$\therefore$ 关于 x 的方程 $x^2-ax+a-1=0$ 有两个相等的实数根, $\therefore (-a)^2-4 \times 1 \times (a-1)=0$, 解得 $a_1=a_2=2,$

$\therefore$ 原方程为 $x^2-2x+1=0$, 即 $(x-1)^2=0, \therefore x_1=x_2=1, \therefore$ 菱形的周长为 $1 \times 4=4.$

14. (1) $-\sqrt{3}$

(2) 解: 当 $(x-1)^2 > x^2$ 时, $x < \frac{1}{2}$, 则 $(x-1)^2=4$, 解得

$x=-1$ 或 $x=3$ (舍去); 当 $(x-1)^2 = x^2$ 时, $x = \frac{1}{2}$, 则

$(x-1)^2 = x^2 = \frac{1}{4} \neq 4$, 舍去; 当 $(x-1)^2 < x^2$ 时, $x > \frac{1}{2}$,

则 $x^2=4$, 解得 $x=2$ 或 $x=-2$ (舍去).

综上所述, x 的值为 -1 或 2 .

作业 9 用一元二次方程解决问题(1)

1. B 2. 12 3. 34 或 -34 4. 10%

5. 解: (1) 设将绿地的长、宽均增加 x m, 则新的矩形绿地的长为 $(35+x)$ m, 宽为 $(15+x)$ m, 根据题意, 得 $(35+x)(15+x)=800,$

整理, 得 $x^2+50x-275=0,$

解得 $x_1=5, x_2=-55$ (不符合题意, 舍去),

所以 $35+x=35+5=40, 15+x=15+5=20.$

答: 新的矩形绿地的长为 40 m, 宽为 20 m.

(2) 设将绿地的长、宽均增加 y m, 则新的矩形绿地的长为 $(35+y)$ m, 宽为 $(15+y)$ m,

根据题意, 得 $(35+y):(15+y)=5:3,$

$$\text{即 } 3(35+y)=5(15+y),$$

解得 $y=15,$

$$\text{所以 } (35+y)(15+y)=(35+15) \times (15+15)=1500.$$

答: 新的矩形绿地的面积为 $1500 \text{ m}^2.$

6. A 7. 10 8. 2

9. 解: (1) 设该市改造小区投入资金的年平均增长率为 $x,$

根据题意, 得 $1000(1+x)^2=1440,$

解得 $x_1=0.2=20\%, x_2=-2.2$ (不合题意, 舍去).

答: 该市改造小区投入资金的年平均增长率为 $20\%.$

$$(2) 1440 \times (1+20\%) \div 80 = 21.6,$$

答: 该市 2025 年最多可以改造 21 个小区.

10. 解: (1) 设边 AB 的长为 x 米, 则 $AD=77-4x+3=(80-4x)$ 米,

根据题意, 得 $x(80-4x)=300$, 解得 $x_1=5, x_2=15.$

$\because$ 墙的最大可用长度为 30 米, 且当 $x=5$ 时, $AD=80-4 \times 5=60$ (米), 不合题意, 舍去,

$\therefore x=15.$

答: 边 AB 的长为 15 米.

(2) 羊圈的总面积不能为 440 平方米, 理由如下:

$$\text{令 } x(80-4x)=440,$$

整理, 得 $x^2-20x+110=0,$

$$\because (-20)^2-4 \times 1 \times 110=-40<0,$$

$\therefore$ 该方程无解,

$\therefore$ 羊圈的总面积不能为 440 平方米.

11. 解: (1) 设 A 社区居民人口有 x 万人, 则 B 社区有 $(7.5-x)$ 万人, 根据题意, 得 $7.5-x \leq 2x$, 解得 $x \geq 2.5.$

答: A 社区居民人口至少有 2.5 万人.

(2) 根据题意, 得 $1.2(1+m\%)^2+1 \times (1+m\%) \times (1+2m\%)=7.5 \times 76\%.$

设 $m\%=a$, 方程可化为

$$1.2(1+a)^2+(1+a)(1+2a)=5.7,$$

化简, 得 $32a^2+54a-35=0,$

解得 $a=0.5$ 或 $a=-\frac{35}{16}$ (不合题意, 舍去), $\therefore m=50.$

答: m 的值为 50.

作业 10 用一元二次方程解决问题(2)

任务 3: 因为两个方案制作的两种无盖纸盒高度相等, 所以底面积大的方案的纸盒的体积就大.

由任务 1 可知方案 1 中纸盒的底面积为 $(40-2x)^2 \text{ cm}^2$.

由任务 2 可知方案 2 中纸盒的底面积为 $20(40-2x) \text{ cm}^2$.

根据题意, 得 $\begin{cases} x > 0, \\ 40-2x > 0, \end{cases}$ 故 $0 < x < 20$.

当 $(40-2x)^2 > 20(40-2x)$ 时, 解得 $0 < x < 10$;

当 $(40-2x)^2 = 20(40-2x)$ 时, 解得 $x = 10$;

当 $(40-2x)^2 < 20(40-2x)$ 时, 解得 $10 < x < 20$.

故当 $0 < x < 10$ 时, 方案 1 的纸盒体积大; 当 $x = 10$ 时, 方案 1 与方案 2 的纸盒体积一样大; 当 $10 < x < 20$ 时, 方案 2 的纸盒体积大.

任务 4: 方案 2 中单个无盖纸盒的体积为 $20(40-2x)x = -40x^2 + 800x = -40(x-10)^2 + 4000$,

当 $x = 10$ 时, 单个无盖纸盒体积的最大值为 4000 cm^3 .

小结与思考

1. C 2. C 3. C 4. C 5. C 6. $-\frac{2}{3}$

7. 0 8. 3

9. (1) $x_1 = x_2 = -\frac{1}{3}$ (2) $x_1 = \sqrt{41} - 4, x_2 = -\sqrt{41} - 4$

(3) 无实数解 (4) $x_1 = 3, x_2 = 1$

10. 证明: (1) $\because b^2 - 4a \times 2 = b^2 - 8a, a < 0$,

$\therefore b^2 - 8a > 0$,

$\therefore$ 该方程总有两个不相等的实数根.

(2) $\because x_1, x_2$ 是方程 $ax^2 + bx + 2 = 0 (a < 0)$ 的两根,

$\therefore x_1 + x_2 = -\frac{b}{a}, x_1 x_2 = \frac{2}{a}$.

$\because x_2 = -2x_1, \therefore x_1 + (-2x_1) = -\frac{b}{a}, x_1(-2x_1) = \frac{2}{a}$,

即 $x_1 = \frac{b}{a}, x_1^2 = -\frac{1}{a}$,

$\therefore \left(\frac{b}{a}\right)^2 = -\frac{1}{a}$, 即 $a + b^2 = 0$.

11. 解: (1) 设 $BC = x \text{ m}$, 则 $AB = (33-3x) \text{ m}$,

根据题意, 得 $x(33-3x) = 90$, 解得 $x_1 = 6, x_2 = 5$.

当 $x = 6$ 时, $33-3x = 15$, 符合题意.

当 $x = 5$ 时, $33-3x = 18, 18 > 15$, 不合题意, 舍去.

答: 养鸡场的长(AB)为 15 m , 宽(BC)为 6 m .

(2) 不能. 理由如下:

设 $BC = y \text{ m}$, 则 $AB = (33-3y) \text{ m}$,

根据题意, 得 $y(33-3y) = 100$,

整理, 得 $3y^2 - 33y + 100 = 0$.

$\because b^2 - 4ac = (-33)^2 - 4 \times 3 \times 100 = -11 < 0$,

$\therefore$ 该方程无实数解, 即该农户不能建成一个 100 m^2 的

矩形养鸡场.

12. 解: 设该产品的质量档次是第 x 档次, 则每天的产量为

$[95-5(x-1)]$ 件, 每件利润是 $[6+2(x-1)]$ 元,

根据题意, 得 $[6+2(x-1)][95-5(x-1)] = 1120$,

整理, 得 $x^2 - 18x + 72 = 0$, 解得 $x_1 = 6, x_2 = 12$ (舍去).

答: 该产品的质量档次为第 6 档次.

13. (1) 证明: $\because m \neq 0, \therefore$ 方程为一元二次方程.

$\because b^2 - 4ac = (4m-2)^2 - 4m(4m-4) = 4 > 0$,

$\therefore$ 方程总有实数根.

(2) ① -2

② 解: $\because x = \frac{-(4m-2) \pm 2}{2m}, \therefore x_1 = -2, x_2 = -2 + \frac{2}{m}$.

$\because$ 当 $m = \pm 1$ 或 $m = \pm 2$ 时, $-2 + \frac{2}{m}$ 为整数,

$\therefore m$ 的值为 $\pm 1, \pm 2$ 时, 方程的两个实数根都是整数.

第 2 章 对称图形——圆

作业 12 圆(1)

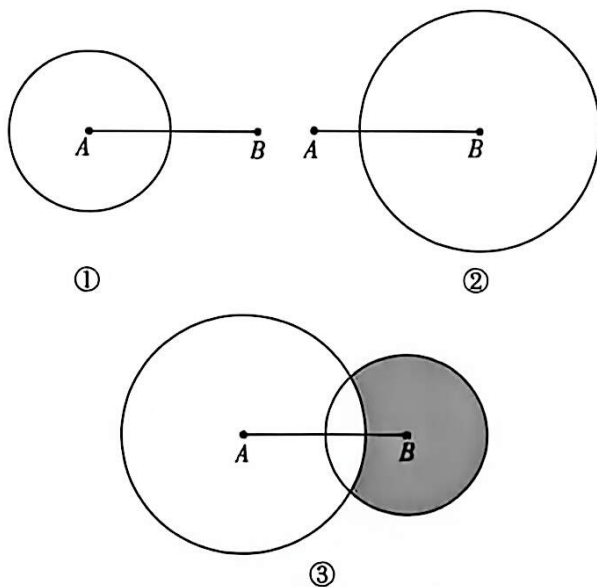
1. D 2. A 3. D

4. $\odot O$ 内 5. < 5 $r > 5$

5. 解: (1) 如答图①, $\odot A$ 即为所作.

(2) 如答图②, $\odot B$ 即为所作.

(3) 如答图③, 阴影部分即为所作.



第 5 题答图

6. B 7. D 8. D 9. $2 \leq a \leq 6$

10. $3 < r < 5$ 11. (3.5, 4) 或 (7.5, 4) 12. $\sqrt{2} + \frac{1}{2}$

13. 证明: (1) 如答图, 连接 AC 交 BD 于点 O , 连接 EO, FO .

$\because$ 四边形 $ABCD$ 是平行四边形, $\therefore O$ 为 AC 的中点.

$\because AE \perp BC, AF \perp CD, \therefore \angle AEC = \angle AFC = 90^\circ$,

$\therefore AO = EO = CO = FO = \frac{1}{2} AC$,

$\because G$ 为 BN 的中点, $\therefore BG=NG$.
 又 $\because \angle BGH=\angle NGM, \therefore \triangle BGH \cong \triangle NGM(SAS)$,
 $\therefore BH=NM, \angle BHG=\angle NMG$,
 则 $BH \parallel NM, \therefore \angle CBH=\angle BFE$.
 $\because$ 四边形 $ABCD$ 是正方形,
 $\therefore AD \parallel BC, \angle ADC=90^\circ$,
 $\therefore \angle BFE=\angle DEM, \angle CDM+\angle EDM=90^\circ$.
 又 $\because DM \perp EF, \therefore \angle DEM+\angle EDM=90^\circ$,
 $\therefore \angle CDM=\angle DEM, \therefore \angle CDM=\angle BFE$,
 $\therefore \angle CBH=\angle CDM$.
 $\because MN=DM, \therefore BH=DM$,
 由正方形的性质可知, $CB=CD$,
 $\therefore \triangle CBH \cong \triangle CDM(SAS)$,
 $\therefore CH=CM, \angle BCH=\angle DCM, \angle BCD=90^\circ$,
 则 $\angle BCH+\angle BCM=\angle DCM+\angle BCM=\angle BCD=90^\circ$,
 $\therefore \triangle MCH$ 是等腰直角三角形.
 $\because HG=MG, \therefore CG \perp MH$, 则 $\triangle CGM$ 也是等腰直角三角
 形, 则 $CG=MG$,
 $\therefore CM=\sqrt{CG^2+MG^2}=\sqrt{2}CG=\sqrt{2}$.

专题(5) 特殊四边形判定中的条件探究问题

1. (1) 解: $AD=\frac{1}{3}BC$.
 理由: $\because AD \parallel BC, DE \parallel AB, AF \parallel DC, AE \parallel DF$,
 $\therefore$ 四边形 $ABED$, 四边形 $AFCD$, 四边形 $AEFD$ 均为平行
 四边形, $\therefore AD=BE=EF=CF, \therefore AD=\frac{1}{3}BC$.
 (2) 解: 当四边形 $ABCD$ 满足条件 $AB=CD$ 时, 四边
 形 $AEFD$ 是矩形.
 理由: $\because AB=CD$, 四边形 $ABED$, 四边形 $AFCD$ 是平行
 四边形, $\therefore AB=DE, AF=CD, \therefore DE=AF$.
 由(1)知四边形 $AEFD$ 是平行四边形,
 $\therefore$ 四边形 $AEFD$ 是矩形.
 (3) $\angle B=\angle C=45^\circ$
2. 解: (1) 四边形 $ADEF$ 是平行四边形.
 理由: $\because \triangle BCE$ 和 $\triangle ABD$ 均为等边三角形,
 $\therefore BE=BC, BD=BA, \angle ABD=60^\circ, \angle CBE=60^\circ$.
 又 $\because \angle DBE=60^\circ-\angle ABE, \angle ABC=60^\circ-\angle ABE$,
 $\therefore \angle DBE=\angle ABC$.
 在 $\triangle BDE$ 和 $\triangle BAC$ 中, $\begin{cases} BE=BC, \\ \angle DBE=\angle ABC, \\ BD=BA, \end{cases}$
 $\therefore \triangle BDE \cong \triangle BAC(SAS), \therefore DE=AC$.
 $\because$ 在等边 $\triangle ACF$ 中, $AC=AF, \therefore DE=AF$.
 同理可得 $DA=EF$,

- $\therefore$ 四边形 $ADEF$ 是平行四边形.
 (2) 当 $\angle BAC=150^\circ$ 时, 四边形 $ADEF$ 是矩形.
 理由: $\because \angle DAF=360^\circ-\angle DAB-\angle BAC-\angle CAF=$
 $90^\circ, \therefore \square ADEF$ 是矩形.
 (3) 当 $AB=AC$ 时, 四边形 $ADEF$ 是菱形.
 理由: $\because AB=AC, \therefore AD=AF, \therefore \square ADEF$ 是菱形.
 (4) 当 $\angle BAC=150^\circ$ 且 $AB=AC$ 或 $\angle ABC=\angle ACB=15^\circ$
 时, 四边形 $ADEF$ 是正方形.
 (5) 当 $\angle BAC=60^\circ$ 时, 以 A, D, E, F 为顶点的四边形不
 存在.

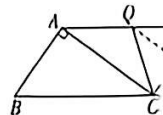
3. (1) 证明: $\because AB \parallel CD$,
 $\therefore \angle CDF=\angle FEB, \angle DCF=\angle EBF$.
 $\because F$ 是 BC 的中点, $\therefore BF=CF$.
 在 $\triangle DCF$ 和 $\triangle EBF$ 中,
 $\begin{cases} \angle CDF=\angle FEB, \\ \angle DCF=\angle EBF, \\ FC=FB, \end{cases}$
 $\therefore \triangle DCF \cong \triangle EBF(AAS), \therefore DC=BE$.
 又 $\because DC \parallel BE, \therefore$ 四边形 $BECD$ 是平行四边形.
 (2) 解: ① $BE=2$.
 理由: 当四边形 $BECD$ 是矩形时, $\angle CEB=90^\circ$.
 $\because \angle ABC=120^\circ, \therefore \angle CBE=60^\circ$,
 $\therefore \angle ECB=30^\circ, \therefore BE=\frac{1}{2}BC=2$.
 ② $BE=4$.
 理由: 当四边形 $BECD$ 是菱形时, $BE=EC$.
 $\because \angle ABC=120^\circ, \therefore \angle CBE=60^\circ$,
 $\therefore \triangle CBE$ 是等边三角形, $\therefore BE=BC=4$.

4. $\frac{4}{3}$ 或 4

5. 解: (1) ① 当点 P 在 BC 上时,
 $DQ=t, PC=10-3t$,
 $\because$ 四边形 $ABCD$ 是平行四边形,
 $\therefore AD \parallel BC, \therefore DQ \parallel PC$.
 若四边形 $PCDQ$ 是平行四边形, 则 $DQ=PC$,
 $\therefore t=10-3t, \therefore t=2.5$;
 ② 当点 P 在 BC 的延长线上时,
 $PC=3t-10$.
 若四边形 $PCDQ$ 是平行四边形,
 则 $DQ=PC, \therefore t=3t-10, \therefore t=5$.
 综上, 当 t 的值为 2.5 或 5 时, 以 P, C, D, Q 为顶点的四
 边形为平行四边形.
 (2) ① 当点 P 在 BC 上时, $DQ=nt, PC=10-nt$,
 若四边形 $PCDQ$ 是菱形,

则 $DQ=PC=CD=AB=6$,
 $\therefore nt=10-nt=6, \therefore nt=4$,
 $\therefore \frac{m}{n}=\frac{2}{3}, \therefore 3m=2n$

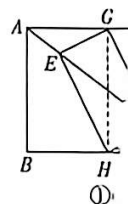
② 如答图, 当点 P 在 BC 的延长线上时, 于点 E .



第 5 题答图

$\because AB \perp AC, \therefore \angle BAC=90^\circ$,
 $\therefore AC=\sqrt{BC^2-AB^2}=\sqrt{10^2-6^2}=8$,
 $\because$ 四边形 $ABCD$ 是平行四边
 $\therefore \angle ACD=90^\circ$,
 $\because$ 四边形 $PCQD$ 是菱形,
 $\therefore PQ \perp CD, CE=DE, PE=$
 $\therefore PQ \parallel AC, \therefore$ 四边形 $ACPF$
 $\therefore PQ=AC=8, \therefore QE=PF$
 $\because CE=DE=\frac{1}{2}CD=\frac{1}{2} \times 10=5$,
 $\therefore DQ=PD=\sqrt{QE^2+DE^2}=\sqrt{5^2+5^2}=5\sqrt{2}$,
 $\therefore nt=mt-10=5, \therefore m=$
 综上, 当 $3m=2n$ 或 $m=$
 边形为菱形.

6. (1) 平行四边形
 (2) 解: 如答图①, 连接
 由(1)得 $AG=BH, AC$
 $\therefore$ 四边形 $ABHG$ 是矩
 ① 如答图①, 当四边
 时, $EF=GH=6$.
 $\because AE=CF=t, \therefore E$.



∵ 四边形 $ALFE$ 是平行四边形.

(2) 当 $\angle BAC = 150^\circ$ 时, 四边形 $ADEF$ 是矩形.

理由: $\because \angle DAF = 360^\circ - \angle DAB - \angle BAC - \angle CAF = 90^\circ$, $\therefore \square ADEF$ 是矩形.

(3) 当 $AB = AC$ 时, 四边形 $ADEF$ 是菱形.

理由: $\because AB = AC$, $\therefore AD = AF$, $\therefore \square ADEF$ 是菱形.

(4) 当 $\angle BAC = 150^\circ$ 且 $AB = AC$ 或 $\angle ABC = \angle ACB = 15^\circ$ 时, 四边形 $ADEF$ 是正方形.

(5) 当 $\angle BAC = 60^\circ$ 时, 以 A, D, E, F 为顶点的四边形不存在.

3. (1) 证明: $\because AB \parallel CD$,

$\therefore \angle CDF = \angle FEB, \angle DCF = \angle EBF$.

$\because F$ 是 BC 的中点, $\therefore BF = CF$.

在 $\triangle DCF$ 和 $\triangle EBF$ 中,

$$\begin{cases} \angle CDF = \angle BEF, \\ \angle DCF = \angle EBF, \\ FC = FB, \end{cases}$$

$\therefore \triangle DCF \cong \triangle EBF (AAS), \therefore DC = BE$.

又 $\because DC \parallel BE$, $\therefore$ 四边形 $BECD$ 是平行四边形.

(2) 解: ① $BE = 2$.

理由: 当四边形 $BECD$ 是矩形时, $\angle CEB = 90^\circ$.

$\because \angle ABC = 120^\circ, \therefore \angle CBE = 60^\circ$,

$\therefore \angle ECB = 30^\circ, \therefore BE = \frac{1}{2} BC = 2$.

② $BE = 4$.

理由: 当四边形 $BECD$ 是菱形时, $BE = EC$.

$\because \angle ABC = 120^\circ, \therefore \angle CBE = 60^\circ$,

$\therefore \triangle CBE$ 是等边三角形, $\therefore BE = BC = 4$.

4. $\frac{4}{3}$ 或 4

5. 解: (1) ① 当点 P 在 BC 上时,

$DQ = t, PC = 10 - 3t$,

$\because$ 四边形 $ABCD$ 是平行四边形,

$\therefore AD \parallel BC, \therefore DQ \parallel PC$.

若四边形 $PCDQ$ 是平行四边形, 则 $DQ = PC$,

$\therefore t = 10 - 3t, \therefore t = 2.5$;

② 当点 P 在 BC 的延长线上时,

$PC = 3t - 10$.

若四边形 $CPDQ$ 是平行四边形,

则 $DQ = PC, \therefore t = 3t - 10, \therefore t = 5$.

综上, 当 t 的值为 2.5 或 5 时, 以 P, C, D, Q 为顶点的四边形为平行四边形.

(2) ① 当点 P 在 BC 上时, $DQ = mt, PC = 10 - mt$,

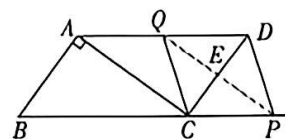
若四边形 $PCDQ$ 是菱形,

则 $DQ = PC = CD = AB = 6$,

$\therefore mt = 10 - mt = 6, \therefore mt = 4$,

$\therefore \frac{m}{n} = \frac{2}{3}, \therefore 3m = 2n$;

② 如答图, 当点 P 在 BC 的延长线上时, 连接 PQ , 交 CD 于点 E .



第 5 题答图

$\because AB \perp AC, \therefore \angle BAC = 90^\circ$,

$\therefore AC = \sqrt{BC^2 - AB^2} = \sqrt{10^2 - 6^2} = 8$.

$\because$ 四边形 $ABCD$ 是平行四边形, $\therefore CD \parallel AB$,

$\therefore \angle ACD = 90^\circ$.

$\because$ 四边形 $PCQD$ 是菱形,

$\therefore PQ \perp CD, CE = DE, PE = QE$,

$\therefore PQ \parallel AC, \therefore$ 四边形 $ACPQ$ 是平行四边形,

$\therefore PQ = AC = 8, \therefore QE = PE = 4$,

$\because CE = DE = \frac{1}{2} CD = \frac{1}{2} AB = 3$,

$\therefore DQ = PD = \sqrt{QE^2 + DE^2} = \sqrt{4^2 + 3^2} = 5$,

$\therefore nt = mt - 10 = 5, \therefore m = 3n$.

综上, 当 $3m = 2n$ 或 $m = 3n$ 时, 以 P, C, D, Q 为顶点的四边形为菱形.

6. (1) 平行四边形

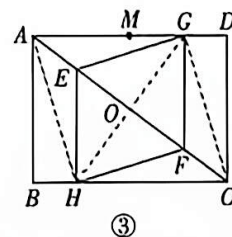
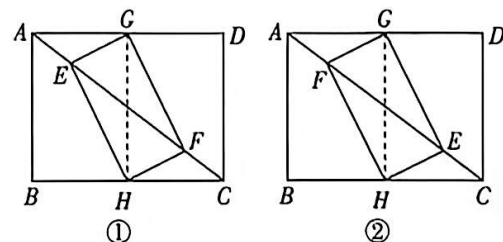
(2) 解: 如答图①, 连接 $GH, AC = \sqrt{AB^2 + BC^2} = 10$,

由(1)得 $AG = BH, AG \parallel BH, \angle B = 90^\circ$,

$\therefore$ 四边形 $ABHG$ 是矩形, $\therefore GH = AB = 6$.

① 如答图①, 当四边形 $EGFH$ 是矩形且点 E 在点 F 左侧时, $EF = GH = 6$.

$\because AE = CF = t, \therefore EF = 10 - 2t = 6, \therefore t = 2$;



第 6 题答图

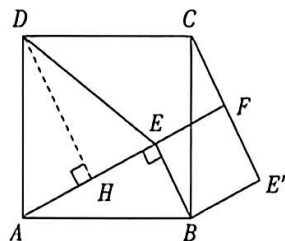
②如答图②,当四边形 $EGFH$ 是矩形且点 E 在点 F 右侧时, $EF=GH=6$,
 $\because AE=CF=t$,
 $\therefore EF=t+t-10=2t-10=6, \therefore t=8$.
 综上,当四边形 $EGFH$ 为矩形时, $t=2$ 或 $t=8$.
 (3)解:如答图③, M 为 AD 的中点,连接 AH, CG, GH , AC 与 GH 交于点 O .
 $\because$ 四边形 $EGFH$ 为菱形,
 $\therefore GH \perp EF, OG=OH, OE=OF$,
 $\therefore OA=OC, AG=AH$,
 $\therefore$ 四边形 $AGCH$ 为菱形, $\therefore AG=CG$.
 设 $AG=CG=x$, 则 $DG=8-x$,
 由勾股定理,得 $CD^2+DG^2=CG^2$,
 即 $6^2+(8-x)^2=x^2$, 解得 $x=\frac{25}{4}$,
 $\therefore MG=\frac{25}{4}-4=\frac{9}{4}$, 即 $t=\frac{9}{4}$.

专题(6) 与正方形有关的旋转问题

1. 解:(1) 四边形 $BE'FE$ 是正方形. 理由如下:

$\because$ 将 $Rt\triangle ABE$ 绕点 B 按顺时针方向旋转 90° 得到 $\triangle CBE'$,
 $\therefore \angle AEB = \angle CE'B = 90^\circ, BE=BE', \angle EBE' = 90^\circ$.
 $\because \angle BEF = 90^\circ, \therefore$ 四边形 $BE'FE$ 是正方形.
 (2) $CF=FE'$. 理由如下:

如答图①,过点 D 作 $DH \perp AE$ 于点 H .



第1题答图①

$\because DA=DE, DH \perp AE$,
 $\therefore AH=\frac{1}{2}AE, \angle ADH + \angle DAH = 90^\circ$.
 $\because$ 四边形 $ABCD$ 是正方形, $\therefore AD=AB, \angle DAB=90^\circ$,
 $\therefore \angle DAH + \angle BAE = 90^\circ, \therefore \angle ADH = \angle BAE$.
 $\because DA=AB, \angle AHD = \angle BEA = 90^\circ$,
 $\therefore \triangle ADH \cong \triangle BAE (AAS)$,
 $\therefore AH=HE=BE=\frac{1}{2}AE$.
 由(1)可知四边形 $BE'FE$ 是正方形, $\therefore BE=E'F$,
 $\therefore E'F=\frac{1}{2}AE$.

$\because$ 将 $Rt\triangle ABE$ 绕点 B 按顺时针方向旋转 90° 得到

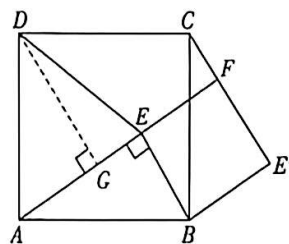
$\triangle CBE'$,

$$\therefore AE=CE', \therefore E'F=\frac{1}{2}CE', \therefore CF=E'F.$$

(3) 作 $DG \perp AE$ 于点 G , 如答图②.

由(2)可知, $Rt\triangle AEB \cong Rt\triangle DGA$,

由将 $Rt\triangle ABE$ 绕点 B 按顺时针方向旋转 90° 得 $Rt\triangle CBE'$ 可知, $Rt\triangle AEB \cong Rt\triangle CE'B$,



第1题答图②

$$\therefore Rt\triangle DGA \cong Rt\triangle CE'B, \therefore DG=AE=CE'.$$

$$\because S_{\triangle ADE} = \frac{1}{2}DG \cdot AE = 72,$$

$$\text{设 } AE=x, \text{ 则 } DG=\frac{144}{x},$$

$$\therefore \text{由 } AE=DG, \text{ 得 } x=\frac{144}{x}, \text{ 解得 } x=12,$$

$$\therefore DG=AE=CE'=12.$$

$$\text{在 } Rt\triangle CBE' \text{ 中, } BE' = \sqrt{BC^2 - CE'^2} = \sqrt{15^2 - 12^2} = 9.$$

$$\because \text{四边形 } BE'FE \text{ 是正方形, } \therefore BE=E'F=9,$$

$$\therefore CF=CE'-E'F=12-9=3.$$

2. (1) 证明: $\because \angle AEF=90^\circ, \therefore \angle AEB + \angle GEF = 90^\circ$.

又 $\because$ 四边形 $ABCD$ 是正方形, $\therefore \angle ABE=90^\circ$,

$$\therefore \angle AEB + \angle BAE = 90^\circ, \therefore \angle GEF = \angle BAE.$$

又 $\because FG \perp BC, \therefore \angle EGF=90^\circ$.

$$\text{在 } \triangle ABE \text{ 和 } \triangle EGF \text{ 中, } \begin{cases} \angle ABE = \angle EGF, \\ \angle BAE = \angle GEF, \\ AE = EF, \end{cases}$$

$$\therefore \triangle ABE \cong \triangle EGF (AAS).$$

(2) 解: 由(1)知 $\triangle ABE \cong \triangle EGF$,

$$\therefore AB=EG, BE=GF.$$

设 $EC=x$, 则 $BE=GF=2EC=2x$.

$$\because S_{\triangle BCF} = 4, \therefore \frac{1}{2}x \cdot 2x = 4, \text{ 解得 } x=2 (\text{负值已舍去}),$$

$$\therefore AB=BC=3x=6.$$

(3) 解: $\triangle FCG$ 是等腰直角三角形. 理由如下:

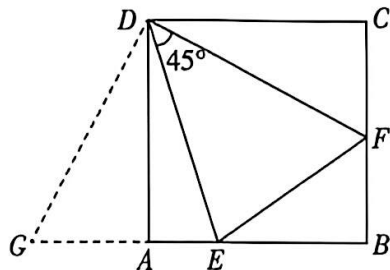
$$\because BC=AB=EG, \therefore BE=CG, \therefore CG=FG.$$

又 $\because \angle EGF=90^\circ, \therefore \triangle FCG$ 是等腰直角三角形.

3. (1) 证明: 如答图, 延长 BA 到点 G , 使 $AG=CF$. 连接 DG .

$\because$ 四边形 $ABCD$ 是正方形,

$$\therefore DA=DC, \angle DAB = \angle DCF = 90^\circ, \therefore \angle DAG = 90^\circ,$$



第3题答图

$\therefore \triangle DAG \cong \triangle DCF$ (SAS),
 $\therefore \angle GDA = \angle FDC, DG = DF$.
 $\because \angle ADC = 90^\circ, \angle EDF = 45^\circ$,
 $\therefore \angle EDG = \angle ADG + \angle ADE = \angle FDC + \angle ADE = 45^\circ$.

在 $\triangle DEG$ 和 $\triangle DEF$ 中, $\begin{cases} DE = DE, \\ \angle EDG = \angle EDF, \\ DG = DF, \end{cases}$

$\therefore \triangle DEG \cong \triangle DEF$ (SAS),
 $\therefore EF = EG = AE + AG = AE + CF$.

(2) ① 4

② $EF = CF - AE$ 或 $EF = AE - CF$

4. 解: (1) $\because$ 四边形 $ABCD$ 是正方形, $\therefore \angle ABC = 90^\circ$.

$\because \triangle ABP \cong \triangle CBQ, \therefore \angle ABP = \angle CBQ$,

$\therefore \angle PBQ = \angle ABC = 90^\circ$.

$\because PB = BQ = 2, \therefore PQ = \sqrt{PB^2 + BQ^2} = \sqrt{8}$.

(2) 由旋转得 $QC = PA = 1$,

在 $\triangle QPC$ 中, $(\sqrt{8})^2 + 1^2 = 3^2$,

即 $PQ^2 + QC^2 = PC^2$,

$\therefore \triangle QPC$ 为直角三角形, $\angle PQC = 90^\circ$.

$\because \triangle PBQ$ 是等腰直角三角形, $\therefore \angle BQP = 45^\circ$,

$\therefore \angle APB = \angle BQC = \angle BQP + \angle PQC = 45^\circ + 90^\circ = 135^\circ$.

周练 2

1. D 2. D 3. B 4. $\sqrt{2}$ 5. 4 6. 7. 5

7. 解: 四边形 $OCDE$ 是菱形, 理由如下:

$\because CD \parallel OE, \therefore \angle FEO = \angle FCD$.

$\because CE$ 是线段 OD 的垂直平分线,

$\therefore FD = FO, ED = OE, CD = CO$,

$\therefore \angle DCF = \angle FCO, \therefore \angle FEO = \angle FCO$,

$\therefore OE = OC, \therefore$ 四边形 $OCDE$ 是菱形.

8. 解: (1) 由题意可得 $BQ = PD = t, AP = QC = 6 - t$.

$\because$ 四边形 $ABCD$ 是矩形, $\therefore \angle B = 90^\circ, AD \parallel BC$,

$\therefore$ 当 $BQ = AP$ 时, 四边形 $ABQP$ 为矩形,

$\therefore \sqrt{3^2 + t^2} = 6 - t$, 解得 $t = \frac{9}{4}, \therefore \frac{9}{4}$

$AQCP$ 是菱形.

作业 22 三角形的中位线

1. C 2. C 3. 4

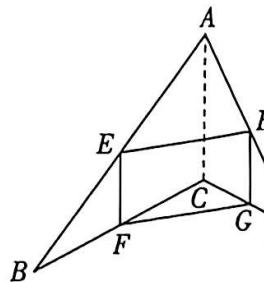
4. (1) 证明: 连接 AC , 如答图.

$\because E, F, G, H$ 分别是 AB, BC, CD, DA

$\therefore EF = \frac{1}{2}AC, EF \parallel AC, HG = \frac{1}{2}AC$.

$\therefore EF = HG, HG \parallel EF$,

$\therefore$ 四边形 $EFGH$ 是平行四边形.



第4题答图

(2) $AC = BD$

5. 4 6. $270^\circ - 3\alpha$ 7. 6. 5

8. (1) 证明: $\because DE$ 和 AF 分别是 $\triangle ABC$

$\therefore AD = \frac{1}{2}AB, EF$ 是 $\triangle ABC$ 的中位

$\therefore EF \parallel AB, EF = \frac{1}{2}AB, \therefore EF = AD$

$\therefore$ 四边形 $ADFE$ 是平行四边形, $\therefore$

(2) 解: 当 $AF = \frac{1}{2}BC$ 时, 四边形 A

理由: $\because$ 线段 DE 为 $\triangle ABC$ 的中位

$\therefore AF = \frac{1}{2}BC, \therefore AF = DE$.

由(1)得, 四边形 $ADFE$ 是平行四

$\therefore$ 四边形 $ADFE$ 为矩形.

9. (1) 证明: $\because \angle ADE = \angle ABC$,

$\therefore DE \parallel BC, \therefore \angle AED = \angle ACB$.

又 $\because \angle EDF = \angle ACB, \therefore \angle AED =$

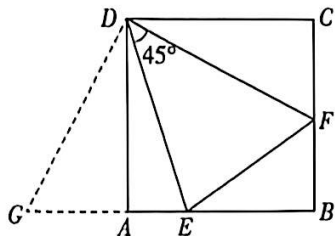
(2) 解: $\because D, E, F$ 分别是 AB, AC

$\therefore DF$ 是 $\triangle ABE$ 的中位线,

$\therefore DF \parallel AE$, 且 $DF = \frac{1}{2}AE$.

设 $S_{\triangle DEF} = x$,

$\because E$ 是 AC 的中点, $\therefore S_{\triangle AEF} = S$



第3题答图

$\therefore \triangle DAG \cong \triangle DCF$ (SAS),
 $\therefore \angle GDA = \angle FDC, DG = DF$.
 $\therefore \angle ADC = 90^\circ, \angle EDF = 45^\circ$,
 $\therefore \angle EDG = \angle ADG + \angle ADE = \angle FDC + \angle ADE = 45^\circ$.

在 $\triangle DEG$ 和 $\triangle DEF$ 中, $\begin{cases} DE = DE, \\ \angle EDG = \angle EDF, \\ DG = DF, \end{cases}$

$\therefore \triangle DEG \cong \triangle DEF$ (SAS),
 $\therefore EF = EG = AE + AG = AE + CF$.

(2) ①4

② $EF = CF - AE$ 或 $EF = AE - CF$

4. 解: (1) $\because$ 四边形 $ABCD$ 是正方形, $\therefore \angle ABC = 90^\circ$.

$\therefore \triangle ABP \cong \triangle CBQ, \therefore \angle ABP = \angle CBQ$,
 $\therefore \angle PBQ = \angle ABC = 90^\circ$.

$\therefore PB = BQ = 2, \therefore PQ = \sqrt{PB^2 + BQ^2} = \sqrt{8}$.

(2) 由旋转得 $QC = PA = 1$,

在 $\triangle QPC$ 中, $(\sqrt{8})^2 + 1^2 = 3^2$,

即 $PQ^2 + QC^2 = PC^2$,

$\therefore \triangle QPC$ 为直角三角形, $\angle PQC = 90^\circ$.

$\therefore \triangle PBQ$ 是等腰直角三角形, $\therefore \angle BQP = 45^\circ$,

$\therefore \angle APB = \angle BQC = \angle BQP + \angle PQC = 45^\circ + 90^\circ = 135^\circ$.

周练 2

1. D 2. D 3. B 4. $\sqrt{2}$ 5. 4 6. 7. 5

7. 解: 四边形 $OCDE$ 是菱形, 理由如下:

$\because CD \parallel OE, \therefore \angle FEO = \angle FCD$.

$\because CE$ 是线段 OD 的垂直平分线,

$\therefore FD = FO, ED = OE, CD = CO$,

$\therefore \angle DCF = \angle FCO, \therefore \angle FEO = \angle FCO$,

$\therefore OE = OC, \therefore$ 四边形 $OCDE$ 是菱形.

8. 解: (1) 由题意可得 $BQ = PD = t, AP = QC = 6 - t$.

$\because$ 四边形 $ABCD$ 是矩形, $\therefore \angle B = 90^\circ, AD \parallel BC$,

$\therefore$ 当 $BQ = AP$ 时, 四边形 $ABQP$ 为矩形,

$\therefore t = 6 - t$, 解得 $t = 3, \therefore$ 当 $t = 3$ 时, 四边形 $ABQP$ 是矩形.

(2) $\because AD \parallel BC, AP = CQ, \therefore$ 四边形 $AQCP$ 为平行四边形, $\therefore$ 当 $AQ = QC$ 时, 四边形 $AQCP$ 是菱形,

$\therefore \sqrt{3^2 + t^2} = 6 - t$, 解得 $t = \frac{9}{4}, \therefore$ 当 $t = \frac{9}{4}$ 时, 四边形

$AQCP$ 是菱形.

作业 22 三角形的中位线

1. C 2. C 3. 4

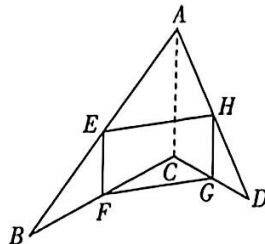
4. (1) 证明: 连接 AC , 如答图.

$\because E, F, G, H$ 分别是 AB, BC, CD, DA 的中点,

$\therefore EF = \frac{1}{2}AC, EF \parallel AC, HG = \frac{1}{2}AC, HG \parallel AC$,

$\therefore EF = HG, EF \parallel HG$,

$\therefore$ 四边形 $EFGH$ 是平行四边形.



第4题答图

(2) $AC = BD$

5. 4 6. $270^\circ - 3\alpha$ 7. 6. 5

8. (1) 证明: $\because DE$ 和 AF 分别是 $\triangle ABC$ 的中位线和中线,

$\therefore AD = \frac{1}{2}AB, EF$ 是 $\triangle ABC$ 的中位线,

$\therefore EF \parallel AB, EF = \frac{1}{2}AB, \therefore EF = AD$,

$\therefore$ 四边形 $ADFE$ 是平行四边形, $\therefore AF$ 与 DE 互相平分.

(2) 解: 当 $AF = \frac{1}{2}BC$ 时, 四边形 $ADFE$ 为矩形.

理由: $\because$ 线段 DE 为 $\triangle ABC$ 的中位线, $\therefore DE = \frac{1}{2}BC$.

$\therefore AF = \frac{1}{2}BC, \therefore AF = DE$.

由(1)得, 四边形 $ADFE$ 是平行四边形,

$\therefore$ 四边形 $ADFE$ 为矩形.

9. (1) 证明: $\because \angle ADE = \angle ABC$,

$\therefore DE \parallel BC, \therefore \angle AED = \angle ACB$.

又 $\because \angle EDF = \angle ACB, \therefore \angle AED = \angle EDF, \therefore DF \parallel AC$.

(2) 解: $\because D, E, F$ 分别是 AB, AC, BE 的中点,

$\therefore DF$ 是 $\triangle ABE$ 的中位线,

$\therefore DF \parallel AE$, 且 $DF = \frac{1}{2}AE$.

设 $S_{\triangle DEF} = x$,

$\because E$ 是 AC 的中点, $\therefore S_{\triangle ADE} = S_{\triangle CEF} = 2x$.

$\because F$ 是 BE 的中点,

$\therefore S_{\triangle BDF} = S_{\triangle DEF} = x, S_{\triangle BCF} = S_{\triangle CEF} = 2x$,

$\therefore S_{\text{四边形}CEFD} = 3x$.

$\therefore S_{\text{四边形}CEFD} = 9, \therefore 3x = 9, \therefore x = 3$,

第2题答图②

$$DQ = DF - QF = \frac{1}{2}CD - 1 = 2 - 1 = 1,$$

$$\therefore CQ = CD - DQ = 4 - 1 = 3, \therefore MQ = CQ = 3.$$

设 $AP = MP = x$,

$$\text{则 } PD = AD - AP = 4 - x, PQ = MP + MQ = x + 3,$$

$$\text{在 Rt}\triangle PDQ \text{ 中, } PD^2 + DQ^2 = PQ^2,$$

$$\therefore (4-x)^2 + 1^2 = (x+3)^2, \text{解得 } x = \frac{4}{7}, \text{即 } AP = \frac{4}{7}.$$

综上, AP 的长为 $\frac{12}{5}$ 或 $\frac{4}{7}$.

小结与思考

1. D 2. A 3. A 4. B

5. 14 6. $\frac{30}{13} \leq AM < 6$ 7. $\frac{5}{3} \leq CF \leq 3$ 8. $\sqrt{3}$

9. (1)解: 四边形 $ACED$ 是平行四边形.

证明: $\because AD \parallel BC, DE \parallel AC,$

$\therefore$ 四边形 $ACED$ 是平行四边形.

(2)证明: 由(1)知四边形 $ACED$ 是平行四边形,

$\therefore AC = DE.$

$\because AC = DB, \therefore DE = DB, \therefore \angle E = \angle DBC.$

$\because DE \parallel AC, \therefore \angle E = \angle ACB, \therefore \angle ACB = \angle DBC.$

又 $\because AC = DB, BC = CB, \therefore \triangle ABC \cong \triangle DCB (SAS),$

$\therefore AB = DC,$

$\therefore$ 梯形 $ABCD$ 是等腰梯形.

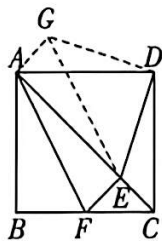
10. 解: $AF = \sqrt{2}DE$

(1)如题图③, 作 $DG \parallel AF$, 交 BC 的延长线于点 G , 连接 EG .

$\because$ 四边形 $ABCD$ 是正方形,
 $\therefore \angle ABC = \angle BCD = 90^\circ, AB = BC = CD = AD,$
 $AD \parallel BC,$
 $\because DG \parallel AF, AD \parallel BC, \therefore$ 四边形 $AFGD$ 为平行四边形,
 $\therefore AF = DG, AD = FG, \therefore FG = CD.$
 $\because \angle ABC = 90^\circ, AB = BC,$
 $\therefore \angle ACB = 45^\circ, \angle ACD = 45^\circ.$
 $\because EF \perp AC, \therefore \angle FEC = 90^\circ, \therefore \angle EFC = \angle ECF = 45^\circ,$
 $\therefore EF = EC, \therefore \angle EFC = \angle ECD,$
 $\therefore \triangle CDE \cong \triangle FGE (SAS),$
 $\therefore ED = EG, \angle FEG = \angle CED,$
 $\therefore \angle DEG = \angle FEC = 90^\circ, \therefore \triangle DEG$ 是等腰直角三角形,
 $\therefore DG^2 = DE^2 + EG^2 = 2DE^2, \therefore DG = \sqrt{2}DE,$
 $\therefore AF = \sqrt{2}DE.$

(2) 如答图, 作 $DG \perp DE$, 并截取 $DG = DE$, 连接 AG ,
 GE , 则 $\triangle DEG$ 是等腰直角三角形,

$\therefore EG^2 = DE^2 + DG^2 = 2DE^2, \therefore EG = \sqrt{2}DE.$
 $\because$ 四边形 $ABCD$ 是正方形, $\therefore \angle ADC = 90^\circ, CD = AD,$
 $\therefore \angle DAC = \angle DCA = 45^\circ$, 同理, $\angle ACB = 45^\circ.$
 $\because \angle ADC = \angle GDE = 90^\circ, \therefore \angle GDA = \angle EDC,$
 $\therefore \triangle GDA \cong \triangle EDC (SAS),$
 $\therefore \angle GAD = \angle ECD = 45^\circ, AG = EC, \therefore \angle GAE = 90^\circ.$
 $\because EF \perp AC, \therefore \angle FEC = \angle FEA = 90^\circ,$
 $\therefore \angle EFC = \angle ECF = 45^\circ, \therefore EF = EC, \therefore EF = AG.$
 $\because \angle GAE = \angle FEA = 90^\circ, \therefore AG \parallel EF,$



第 10 题答图

$\therefore$ 四边形 $AGEF$ 为平行四边形,
 $\therefore AF = EG, \therefore AF = \sqrt{2}DE.$

第 9 章 因式分解

作业 24 因式分解的概念

9. 解: (1) $\because 8a^3b^2 - 12ab^4 + 4ab = 4ab(2a^2b - 3b^3 + 1)$
 $\therefore$ 原式错误, 原因是另一个因式漏项了.
 (2) $\because 4x^4 - 2x^3y = 2x^3(2x - y),$
 $\therefore$ 原式错误, 原因是因式分解不完全.

10. 解: $\because x^2 + mx - 15$ 可以分解为 $(x+3)(x+n)$
 $\therefore x^2 + nx + 3x + 3n = x^2 + mx - 15,$
 $\therefore \begin{cases} m = n + 3, \\ 3n = -15, \end{cases} \therefore \begin{cases} m = -2, \\ n = -5. \end{cases}$

11. (1) -2 (2) 2 1

(3) 解: 设另一个因式为 $(x+n)$, 得 $x^2 -$
 $(x+3)(x+n),$

则 $x^2 - 4x + m = x^2 + (3+n)x + 3n,$

$\therefore \begin{cases} 3+n = -4, \\ 3n = m, \end{cases} \text{ 解得 } \begin{cases} n = -7, \\ m = -21, \end{cases}$

$\therefore$ 另一个因式为 $(x-7)$, m 的值为 -21.

作业 25 提公因式法

1. C 2. (1) $2a$ (2) m^2 (3) ab 3. $x(x-3)$

4. (1) $x(2x+1)$

(2) $(x-y)(3m+2n)$

(3) $(a-1)(a+1)$

(4) $4(a-b)^2$

5. A 6. A 7. $a(a+5)$ 8. 0

9. (1) $x(a-1)^2$

(2) $2(b-1)^2(3a-1)$

(3) $x(x-y)(x-4)$ (4) $2a(a+b)$

10. (1) 3. 98 (2) -2019

11. 解: (1) $\because m+n=2, mn=-3,$

$\therefore$ 原式 $= 2 - mn(m+n) = 2 - (-3) \times 2 = 8.$

(2) $\because a+b=3, ab=1,$

$\therefore a^2b + ab^2 = ab(a+b) = 1 \times 3 = 3.$

12. 解: $\because a$ 是一个正整数, 且 a 除以 3 余 1,

$\therefore$ 设 $a = 3x+1$ (x 是非负整数),

则 $a^2 + 4a + 4 = (a+2)^2 = (3x+1+2)^2 =$
 $9(x+1)^2.$

$\because (x+1)^2$ 是正整数, $\therefore 9(x+1)^2$ 能被 9 整

$\therefore a^2 + 4a + 4$ 能被 9 整除.

13. (1) 提公因式法 2

$$\therefore CD:BD=DE:CD.$$

$$\therefore \angle CDE = \angle BDC, \therefore \triangle CDE \sim \triangle BDC,$$

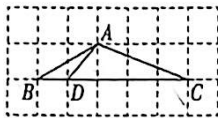
$$\therefore \angle CBD = \angle ECD.$$

训练(2) 三角形相似的判定

1. D 2. B 3. A 4. B 5. 1.2 6. 2 7. 2 8. 3

9. 15 10. $\frac{1+\sqrt{17}}{2}$

11. (1) 解: 如答图, D 是所作点.



第 11 题答图

(2) 证明: $\because AB = \sqrt{1^2 + 2^2} = \sqrt{5}, BC = 5, BD = 1,$

$$\therefore \frac{BD}{AB} = \frac{1}{\sqrt{5}} = \frac{\sqrt{5}}{5}, \frac{AB}{BC} = \frac{\sqrt{5}}{5}, \therefore \frac{BD}{AB} = \frac{AB}{BC}.$$

$$\therefore \angle DBA = \angle ABC, \therefore \triangle ABD \sim \triangle CBA.$$

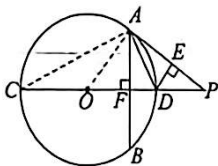
12. (1) 证明: 连接 OA, 如答图.

$$\because AB \perp CD, \therefore \angle AFD = 90^\circ, \therefore \angle FAD + \angle ADF = 90^\circ.$$

$$\because OA = OD, \therefore \angle OAD = \angle ADF, \therefore \angle FAD + \angle OAD = 90^\circ.$$

$$\because \angle EAD = \angle FAD, \therefore \angle EAD + \angle OAD = 90^\circ, \text{ 即 } \angle OAE = 90^\circ, \therefore OA \perp AE.$$

$$\because OA \text{ 是 } \odot O \text{ 的半径}, \therefore AE \text{ 是 } \odot O \text{ 的切线}.$$



第 12 题答图

(2) 解: 连接 AC, 如答图.

$$\because CD \text{ 为 } \odot O \text{ 的直径}, \therefore \angle CAD = 90^\circ,$$

$$\therefore \angle C + \angle ADC = 90^\circ.$$

$$\because \angle FAD + \angle ADC = 90^\circ, \therefore \angle C = \angle FAD.$$

$$\because \angle EAD = \angle FAD, \therefore \angle C = \angle EAD.$$

$$\because \angle P = \angle P, \therefore \triangle ADP \sim \triangle CAP, \therefore \frac{AP}{CP} = \frac{PD}{AP}.$$

$$\because PA = 4, PD = 2, \therefore \frac{4}{CP} = \frac{2}{4}, \text{ 解得 } CP = 8,$$

$$\therefore CD = CP - PD = 8 - 2 = 6, \therefore \odot O \text{ 的半径为 } 3.$$

$$\because OA = OD = 3, \therefore OP = OD + PD = 5.$$

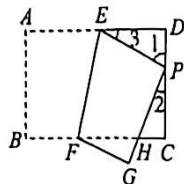
$$\because DE \perp AP, \therefore \angle DEP = 90^\circ,$$

$$\therefore \angle OAP = \angle DEP, \text{ 又 } \angle P = \angle P, \therefore \triangle OAP \sim \triangle DEP,$$

$$\therefore \frac{DE}{OA} = \frac{PD}{OP}, \text{ 即 } \frac{DE}{3} = \frac{2}{5}, \therefore DE = \frac{6}{5},$$

$$\therefore \odot O \text{ 的半径为 } 3, DE \text{ 的长为 } \frac{6}{5}.$$

13. (1) 证明: 如答图①,



第 13 题答图①

$$\because \text{四边形 } ABCD \text{ 是矩形}, \therefore \angle A = \angle D = \angle C = 90^\circ,$$

$$\therefore \angle 1 + \angle 3 = 90^\circ.$$

$$\because \text{点 } E, F \text{ 分别在 } AD, BC \text{ 上, 将矩形 } ABCD \text{ 沿 } EF \text{ 折叠, 使点 } A \text{ 的对应点 } P \text{ 落在 } DC \text{ 上,}$$

$$\therefore \angle EPD = \angle A = 90^\circ, \therefore \angle 1 + \angle 2 = 90^\circ, \therefore \angle 3 = \angle 2, \therefore \triangle DEP \sim \triangle CPH.$$

(2) 解: $\because \text{四边形 } ABCD \text{ 是矩形}, \therefore CD = AB = 2, AD = BC = 3, \angle A = \angle D = \angle C = 90^\circ.$

$$\because P \text{ 为 } CD \text{ 的中点}, \therefore DP = CP = \frac{1}{2} \times 2 = 1.$$

$$\text{设 } EP = AE = x, \therefore ED = 3 - x.$$

$$\text{在 } Rt\triangle EDP \text{ 中}, EP^2 = ED^2 + DP^2, \text{ 即 } x^2 = (3-x)^2 + 1,$$

$$\text{解得 } x = \frac{5}{3},$$

$$\therefore EP = AE = x = \frac{5}{3}, \therefore ED = 3 - \frac{5}{3} = \frac{4}{3}.$$

$$\because \triangle DEP \sim \triangle CPH, \therefore \frac{ED}{PC} = \frac{EP}{PH}, \text{ 即 } \frac{\frac{4}{3}}{1} = \frac{\frac{5}{3}}{PH},$$

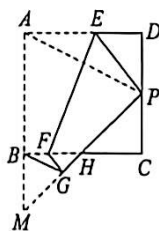
$$\therefore PH = \frac{5}{4}.$$

$$\text{由折叠的性质得 } PG = AB = 2,$$

$$\therefore GH = PG - PH = 2 - \frac{5}{4} = \frac{3}{4}.$$

(3) 解: $AB = \sqrt{6} BG$. 理由如下:

如答图②, 延长 AB, PG 交于点 M, 连接 AP.



第 13 题答图②

$$\because \text{点 } E, F \text{ 分别在 } AD, BC \text{ 上, 将矩形 } ABCD \text{ 沿 } EF \text{ 折叠, 使点 } A \text{ 的对应点 } P \text{ 落在 } CD \text{ 上,}$$

$$\therefore AP \perp EF, BG \perp \text{直线 } EF, \therefore BG \parallel AP.$$

$$\because AE = EP, \therefore \angle EAP = \angle EPA, \therefore \angle BAP = \angle GPA,$$

$$\therefore \triangle MAP \text{ 是等腰三角形}, \therefore MA = MP.$$

$$\because P \text{ 为 } CD \text{ 的中点}, \therefore \text{设 } DP = CP = y,$$

$$\therefore AB = PG = CD = 2y.$$

$$\because H \text{ 为 } BC \text{ 的中点},$$

$$\therefore \angle BHM = \angle CHP$$

$$\therefore \triangle MBH \cong \triangle PCH$$

$$\therefore BM = CP = y, H$$

$$3y, \therefore HP = \frac{1}{2} PA$$

$$\text{在 } Rt\triangle PCH \text{ 中},$$

$$\therefore BC = 2CH = \sqrt{2}$$

$$\text{在 } Rt\triangle APD \text{ 中}$$

$$\therefore BG \parallel AP, \therefore$$

$$\therefore BG = \frac{\sqrt{6}}{3} y,$$

作业 20 相

1. A 2. C 3.

5. (1) 证明: $\because$

$$\therefore \angle BAC =$$

$$\text{即 } \angle BAE =$$

$$\therefore \angle ABE =$$

$$(2) \text{ 解: } \therefore$$

$$\therefore \triangle AOE$$

$$\therefore S_{\triangle AOE}$$

6. D 7. C

10. 解: (1

$$(2)$$

$$\therefore$$

$$\therefore$$

$$\therefore$$

$$\therefore$$

11.

是矩形, $\therefore CD=AB=2, AD=3=90^\circ$.

$$P=CP=\frac{1}{2}\times 2=1.$$

$$3-x.$$

$$ED^2+DP^2, \text{ 即 } x^2=(3-x)^2+1,$$

$$ED=3-\frac{5}{3}=\frac{4}{3}.$$

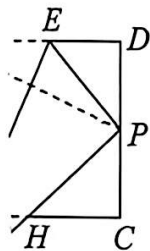
$$\frac{D}{C}=\frac{EP}{PH}, \text{ 即 } \frac{\frac{4}{3}}{1}=\frac{\frac{5}{3}}{PH},$$

$$3=2,$$

$$=\frac{3}{4}.$$

下:

于点 M , 连接 AP .



题答图②

上, 将矩形 $ABCD$ 沿 EF 折
在 CD 上,

$$\therefore BG \parallel AP.$$

$$\angle EPA, \therefore \angle BAP = \angle GPA,$$

$$MA = MP.$$

$$CP = y,$$

作业 20 相似三角形的性质(1)

1. A 2. C 3. D 4. $2:3$

5. (1) 证明: $\because \angle BAC = \angle EAF,$

$$\therefore \angle BAC - \angle EAC = \angle EAF - \angle EAC,$$

$$\text{即 } \angle BAE = \angle CAF.$$

$$\because \angle ABE = \angle ACF, \therefore \triangle ABE \sim \triangle ACF.$$

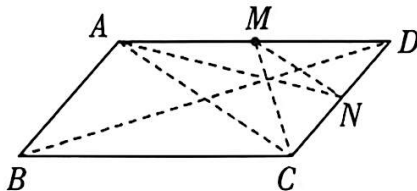
(2) 解: $\because \angle ABE = \angle ACF, \angle AOB = \angle FOC,$

$$\therefore \triangle AOB \sim \triangle FOC, \therefore \frac{S_{\triangle AOB}}{S_{\triangle FOC}} = \left(\frac{AB}{CF}\right)^2 = \frac{9}{16}.$$

$$\because S_{\triangle AOB} = 9, \therefore S_{\triangle FOC} = 16.$$

6. D 7. C 8. $\frac{1}{3}$ 9. $\frac{5}{3}$

10. 解: (1) 如答图, 点 N 即为所求.



第 10 题答图

(2) 连接 MN , 如答图, $\because AM = DM, DN = CN,$

$$\therefore MN \parallel AC, AC = 2MN, \therefore \triangle DMN \sim \triangle DAC,$$

$$\therefore \frac{S_{\triangle DMN}}{S_{\triangle DAC}} = \left(\frac{MN}{AC}\right)^2 = \frac{1}{4},$$

$$\therefore S_{\triangle DAC} = 4 \times 8 = 32,$$

$$\therefore S_{\square ABCD} = 2S_{\triangle DAC} = 64.$$

11. 解: (1) $\because$ 四边形 $BFED$ 是平行四边形,

$$\therefore DE \parallel BC, \therefore \triangle ADE \sim \triangle ABC,$$

$$\therefore \frac{AD}{AB} = \frac{DE}{BC} = \frac{1}{4}.$$

$$\because AB = 8, \therefore AD = 2.$$

(2) $\because \triangle ADE \sim \triangle ABC,$

$$\therefore \frac{S_{\triangle ADE}}{S_{\triangle ABC}} = \left(\frac{DE}{BC}\right)^2 = \left(\frac{1}{4}\right)^2 = \frac{1}{16}.$$

$\because \triangle ADE$ 的面积为 1, $\therefore \triangle ABC$ 的面积是 16.

$\because$ 四边形 $BFED$ 是平行四边形,

$\therefore DE=BF, EF \parallel AB, \therefore \triangle EFC \sim \triangle ABC$.

$\therefore \frac{DE}{BC} = \frac{1}{4}, BC=BF+CF=DE+CF, \therefore \frac{CF}{BC} = \frac{3}{4},$

$\therefore \frac{S_{\triangle EFC}}{S_{\triangle ABC}} = \left(\frac{3}{4}\right)^2 = \frac{9}{16}, \therefore S_{\triangle EFC} = 9,$

$\therefore S_{\square BFED} = S_{\triangle ABC} - S_{\triangle EFC} - S_{\triangle ADE} = 16 - 9 - 1 = 6.$

12. 解: $\because \triangle ABC$ 与 $\triangle DEC$ 的面积相等,

$\therefore \triangle CDF$ 与四边形 $AFEB$ 的面积相等.

$\because AB \parallel DE, \therefore \triangle CEF \sim \triangle CBA$.

$\because EF=9, AB=12, \therefore EF:AB=9:12=3:4,$

$\therefore S_{\triangle CEF}:S_{\triangle CBA}=9:16.$

设 $\triangle CEF$ 的面积为 $9k$, 则 $\triangle CBA$ 的面积为 $16k$, 故四边形 $AFEB$ 的面积为 $7k$.

$\because \triangle CDF$ 与四边形 $AFEB$ 的面积相等, $\therefore S_{\triangle CDF}=7k.$

$\because \triangle CDF$ 与 $\triangle CEF$ 是同高不同底的三角形,

$\therefore S_{\triangle CDF}:S_{\triangle CEF}=DF:EF,$

$\therefore DF:EF=7k:9k=7:9. \because EF=9, \therefore DF=7.$

作业 21 相似三角形的性质(2)

1. C 2. D 3. $5:4$ 4. $\frac{4}{25}$

5. 解: $\because \angle CDE + \angle B = 180^\circ, \angle ADE + \angle CDE = 180^\circ,$

$\therefore \angle ADE = \angle B.$

$\because \angle EAD = \angle CAB, \therefore \triangle EAD \sim \triangle CAB,$

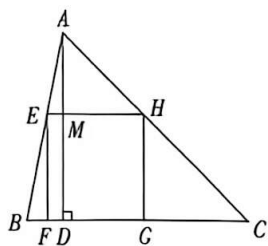
$\therefore \frac{AD}{AB} = \frac{AF}{AG}, \therefore \frac{3}{5} = \frac{AF}{4},$ 解得 $AF = \frac{12}{5}.$

6. A 7. C 8. 2 9. $\frac{24}{5}$

10. (1) 证明: $\because$ 四边形 $EFGH$ 是正方形, $\therefore EH \parallel BC,$

$\therefore \angle AEH = \angle B, \angle AHE = \angle C, \therefore \triangle AEH \sim \triangle ABC.$

(2) 解: 如答图, 设 AD 与 EH 交于点 $M.$



第 10 题答图

$\because \angle EFD = \angle FEM = \angle FDM = 90^\circ,$

$\therefore$ 四边形 $EFDM$ 是矩形, $\therefore EF = DM.$

设正方形 $EFGH$ 的边长为 x cm,

$\because \triangle AEH \sim \triangle ABC, \therefore \frac{EH}{BC} = \frac{AM}{AD},$

$\therefore \frac{x}{40} = \frac{30-x}{30},$ 解得 $x = \frac{120}{7},$

$\therefore$ 正方形 $EFGH$ 的边长为 $\frac{120}{7}$ cm, 面积为 $\frac{14400}{49}$ cm².

11. (1) 证明: $\because AD$ 是 $\triangle ABC$ 的角平分线, $\therefore \angle BAD \approx$

$\angle DAC. \because \angle E = \angle BAD, \therefore \angle E = \angle DAC.$

$\because BE \parallel AD, \therefore \angle E = \angle EDA, \therefore \angle EDA = \angle DAC,$

$\therefore ED \parallel AC.$

(2) 解: $\because BE \parallel AD, \therefore \angle EBD = \angle ADC.$

$\because \angle E = \angle DAC, \therefore \triangle EBD \sim \triangle ADC,$ 且相似比 $k = \frac{BD}{DC} = 2,$

$\therefore \frac{S_1}{S_2} = k^2 = 4,$ 即 $S_1 = 4S_2.$

$\because S_1^2 - 16S_2 + 4 = 0, \therefore 16S_2^2 - 16S_2 + 4 = 0,$

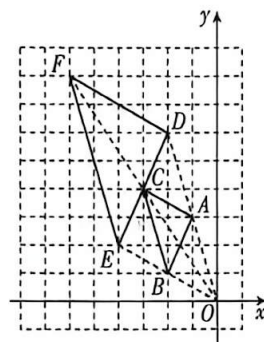
即 $(4S_2 - 2)^2 = 0,$ 解得 $S_2 = \frac{1}{2}.$

$\therefore \frac{S_{\triangle ABC}}{S_2} = \frac{BC}{CD} = \frac{BD+CD}{CD} = \frac{3CD}{CD} = 3, \therefore S_{\triangle ABC} = \frac{3}{2}.$

作业 22 图形的位似

1. A 2. $1:3$ 3. $(4,4)$ 或 $(-4,-4)$

4. (1) 解: 如答图, $\triangle DEF$ 即为所求.



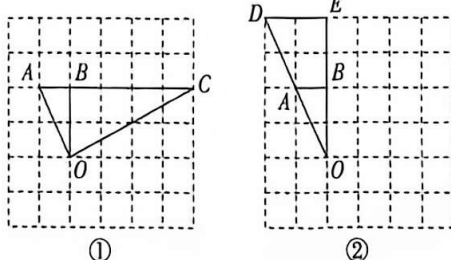
第 4 题答图

(2) $(-2,6)$

(3) $1:3$

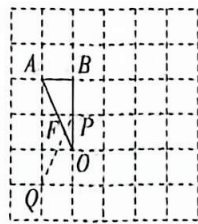
5. A 6. A 7. $(6-2a, -2b)$

8. 解: (1) 如答图①, $\triangle OBC$ 即为所求.



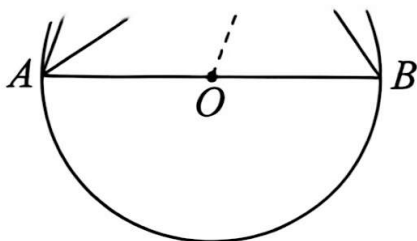
①

②



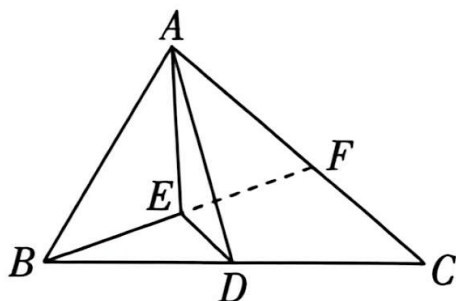
③

第 8 题答图



第 11 题答图

12. 证明: 如答图, 延长 BE 交 AC 于点 F .



第 12 题答图

$$\because \angle ABE = \angle C, \angle BAE = \angle CAD,$$

$$\therefore \triangle ABE \sim \triangle ACD, \therefore \frac{BE}{CD} = \frac{AE}{AD}.$$

$$\because \angle BAE = \angle CAD, \therefore \angle BAE + \angle DAE = \angle CAD + \angle DAE, \therefore \angle BAD = \angle FAE.$$

$$\because \angle AEF = \angle ABE + \angle BAE, \angle ADB = \angle CAD + \angle C,$$

$$\therefore \angle AEF = \angle ADB,$$

$$\therefore \triangle EAF \sim \triangle DAB, \therefore \frac{EF}{BD} = \frac{AE}{AD}, \therefore \frac{BE}{CD} = \frac{EF}{BD}.$$

$$\because D \text{ 是 } BC \text{ 的中点}, \therefore CD = BD, \therefore BE = EF,$$

$$\therefore DE \parallel AC.$$

专题(6) 相似三角形的基本模型

1. D 2. $1:8$

3. (1) 证明: $\because \angle AED = \angle B, \angle BAC = \angle EAD,$

$$\therefore \angle ADE = \angle C.$$

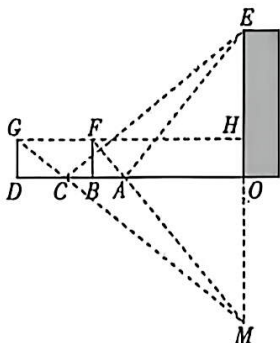
$$\because EF=0.1 \text{ 米}, DF=0.2 \text{ 米}, \therefore DF=2EF,$$

$$\therefore MQ=\frac{1}{2}DQ=\frac{1}{2}\times 23.6=11.8(\text{米}),$$

$$\therefore MN=MQ+QG-GN=11.8+1.5-12=1.3(\text{米}).$$

答:旗帜的宽度 MN 是 1.3 米.

9. 解:如答图,设点 E 关于 AO 的对称点为 M ,由光的反射定律知,延长 GC , FA 相交于点 M ,连接 GF 并延长交 OE 于点 H .



第 9 题答图

$$\because GF \parallel AC, \therefore \triangle MAC \sim \triangle MFG, \triangle MAO \sim \triangle MFH,$$

$$\therefore \frac{AC}{FG} = \frac{MA}{MF} = \frac{MO}{MH},$$

$$\text{即 } \frac{AC}{BD} = \frac{OE}{MH} = \frac{OE}{MO+OH} = \frac{OE}{OE+BF},$$

$$\therefore \frac{OE}{OE+1.6} = \frac{2}{2.1}, \text{解得 } OE=32 \text{ m}.$$

答:楼的高度 OE 为 32 m.

专题(7) 特殊图形中的相似

1. 解:(1)由题意,得 $AQ=2t$ cm, $BP=t$ cm, 则 $AP=(3-t)$ cm.

$$\because \angle ABC=90^\circ, AB=3 \text{ cm}, BC=4 \text{ cm},$$

$$\therefore AC=\sqrt{AB^2+BC^2}=5 \text{ cm}.$$

$$\because \angle A=\angle A,$$

$$\therefore \text{当 } \frac{AQ}{AC} = \frac{AP}{AB} \text{ 时, } \triangle AQP \sim \triangle ACB, \text{ 即 } \frac{2t}{5} = \frac{3-t}{3},$$

$$\text{解得 } t=\frac{15}{11};$$

$$\text{当 } \frac{AQ}{AB} = \frac{AP}{AC} \text{ 时, } \triangle AQP \sim \triangle ABC, \text{ 即 } \frac{2t}{3} = \frac{3-t}{5},$$

$$\text{解得 } t=\frac{9}{13}.$$

$$\therefore \text{当 } t \text{ 的值为 } \frac{15}{11} \text{ 或 } \frac{9}{13} \text{ 时, } \triangle APQ \text{ 与 } \triangle ABC \text{ 相似}.$$

$$(2) \text{ 当 } AQ=AP \text{ 时, } 2t=3-t, \text{ 解得 } t=1;$$

当 $PA=PQ$ 时,作 $PM \perp AQ$ 于点 M ,如答图①, 则 $AM=MQ=t$ cm.

$$\because \angle MAP=\angle BAC, \therefore \triangle AMP \sim \triangle ABC,$$

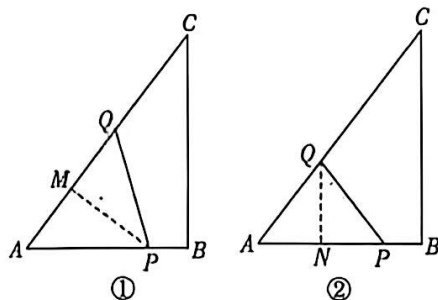
$$\therefore \frac{AM}{AB} = \frac{AP}{AC}, \text{ 即 } \frac{t}{3} = \frac{3-t}{5}, \text{ 解得 } t=\frac{9}{8};$$

当 $QA=QP$ 时,作 $QN \perp AP$ 于点 N ,如答图②, 则 $AN=$

$$PN=\frac{1}{2}(3-t) \text{ cm}, QN \parallel BC, \therefore \triangle ANQ \sim \triangle ABC,$$

$$\therefore \frac{AQ}{AC} = \frac{AN}{AB}, \text{ 即 } \frac{2t}{5} = \frac{\frac{1}{2}(3-t)}{3}, \text{ 解得 } t=\frac{15}{17}.$$

$\therefore$ 当 t 的值为 1 或 $\frac{9}{8}$ 或 $\frac{15}{17}$ 时, $\triangle APQ$ 为等腰三角形.



第 1 题答图

2. (1) 证明:如答图, $\because \angle ADC = \angle 1 + \angle 2 = \angle B + \angle 3$, $\angle 1 = \angle B$,

$$\therefore \angle 2 = \angle 3.$$

$$\text{又 } \because AB=AC, \therefore \angle B = \angle C, \therefore \triangle DCE \sim \triangle ABD.$$

$$(2) \text{ 解: } \because \triangle DCE \sim \triangle ABD, \therefore \frac{CE}{BD} = \frac{DC}{AB}, \text{ 即 } \frac{CE}{x} = \frac{8-x}{6},$$

$$\therefore CE = -\frac{1}{6}x^2 + \frac{4}{3}x.$$

$$\because CE=2, \therefore -\frac{1}{6}x^2 + \frac{4}{3}x = 2, \text{ 解得 } x_1=2, x_2=6.$$

$$(3) \text{ 解: } \textcircled{1} \text{ 当 } DA=DE \text{ 时, } \triangle DCE \cong \triangle ABD, \therefore DC=AB=6, \text{ 即 } 8-x=6, \text{ 解得 } x=2;$$

$$\textcircled{2} \text{ 当 } EA=ED \text{ 时, } \angle DAE = \angle 1 = \angle B = \angle C,$$

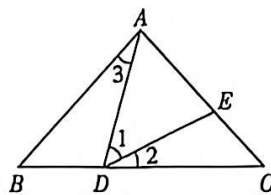
$$\therefore \triangle DAC \sim \triangle ABC, \therefore \frac{DC}{AC} = \frac{AC}{BC}, \text{ 即 } \frac{8-x}{6} = \frac{6}{8},$$

$$\text{解得 } x=\frac{7}{2};$$

$$\textcircled{3} \text{ 当 } AD=AE \text{ 时, } \angle 1 = \angle AED > \angle C,$$

$$\because \angle 1 = \angle B = \angle C, \therefore AD=AE \text{ 的情况不成立}.$$

综上所述,当 $x=2$ 或 $x=\frac{7}{2}$ 时, $\triangle ADE$ 为等腰三角形.



第 2 题答图

3. (1) 证明: $\because \triangle ABC$ 是等边三角形,

$$\therefore \angle A = \angle B = \angle C = \angle EDF = 60^\circ,$$

$$\therefore \angle ADE + \angle BDF = \angle BFD + \angle BDF = 120^\circ,$$

$$\therefore \angle ADE = \angle BFD, \therefore \triangle ADE \sim \triangle BFD.$$

$$(2) \text{ 解: } \because \triangle ADE \sim \triangle BFD, \therefore \frac{AD}{BF} = \frac{AE}{BD} = \frac{DE}{DF}, \text{ 设 } \frac{AD}{BF} =$$

$\frac{AE}{BD} = \frac{DE}{DF} = k$, 则 $\frac{CE}{CF} = \frac{DE}{DF} = k$, 设 $CF = x$, 则 $CE = kx$, 由 $AD \parallel DB = 1:2$, 可设 $AD = 1, DB = 2, \therefore AE = 3 - kx, BF = 3 - x$, $\therefore \frac{1}{3-x} = \frac{3-kx}{2} = k$, 解得 $k = \frac{4}{5}$, 即 $\frac{CE}{CF} = \frac{4}{5}$.

4. 证明: (1) $\because$ 四边形 $ABCD$ 是平行四边形,

$$\therefore \angle B = \angle D.$$

$$\because AM \perp BC, AN \perp CD,$$

$$\therefore \angle AMB = \angle AND = 90^\circ,$$

$$\therefore \triangle AMB \sim \triangle AND.$$

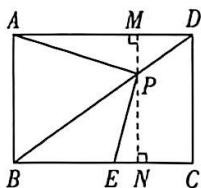
$$(2) \because \triangle AMB \sim \triangle AND, \therefore \frac{AM}{AN} = \frac{AB}{AD}, \angle BAM = \angle DAN.$$

$$\text{又} \because AD = BC, \therefore \frac{AM}{AN} = \frac{AB}{BC}.$$

$$\therefore \angle B + \angle BCD = \angle MAN + \angle BCD = 180^\circ,$$

$$\therefore \angle B = \angle MAN, \therefore \triangle AMN \sim \triangle BAC, \therefore \frac{AM}{AB} = \frac{MN}{AC}.$$

5. 解: 如答图, 过点 P 作 AD 的垂线 MN , 交 AD 于点 M , 交 BC 于点 N . $\because \angle APM + \angle NPE = 180^\circ - \angle APE = 90^\circ, \angle APM + \angle PAM = 180^\circ - \angle AMP = 90^\circ,$



第5题答图

$$\therefore \angle NPE = \angle PAM.$$

$$\text{又} \because \angle AMP = \angle PNE = 90^\circ,$$

$$\therefore \triangle APM \sim \triangle PEN, \therefore \frac{PA}{PE} = \frac{AM}{PN}.$$

$$\because \angle PNB = \angle DCB = 90^\circ, \angle PBN = \angle DBC,$$

$$\therefore \triangle BPN \sim \triangle BDC, \therefore \frac{BN}{PN} = \frac{BC}{CD}.$$

$\because$ 四边形 $AMNB$, 四边形 $ADCB$ 均为矩形,

$$\therefore AM = BN, AD = BC,$$

$$\therefore \frac{AM}{PN} = \frac{AD}{CD} = \frac{10}{6} = \frac{5}{3}, \therefore \frac{PA}{PE} = \frac{5}{3}.$$

6. (1) 证明: $\because$ 四边形 $ABCD$ 是正方形,

$$\therefore AB = AD = CB = CD, \angle ABC = \angle ADC = 90^\circ,$$

$$\therefore \angle BAC = \angle BCA = \angle DAC = \angle DCA = 45^\circ.$$

$$\text{在} \triangle ABE \text{ 和 } \triangle ADE \text{ 中, } \begin{cases} AB = AD, \\ \angle BAE = \angle DAE, \\ AE = AE, \end{cases}$$

$$\therefore \triangle ABE \cong \triangle ADE (\text{SAS}).$$

(2) 解: $\triangle FBG$ 是等腰三角形, 理由如下:

$$\because \triangle ABE \cong \triangle ADE,$$

$$\therefore \angle ABE = \angle ADE, \therefore \angle ABC - \angle ABE = \angle ADC - \angle ADE, \therefore \angle EBC = \angle EDC.$$

$$\because AB \parallel CD, \therefore \angle FGB = \angle EDC, \therefore \angle FGB = \angle EBC.$$

$$\because BF \perp BE, \therefore \angle FBE = 90^\circ, \therefore \angle FGB = \angle EBC = 90^\circ - \angle ABE, \therefore \angle FGB = \angle FBG, \therefore BF = GF, \therefore \triangle FBG \text{ 是等}$$

腰三角形.

$$(3) \text{解: } \because BE = BF = 2, \angle FBE = 90^\circ, \therefore \angle F = \angle BEF =$$

$$45^\circ, \therefore \angle BAC = \angle F, \therefore \angle AEG = \angle FBG.$$

$$\because \angle AGE = \angle FGB \text{ 且 } \angle FGB = \angle FBG,$$

$$\therefore \angle AGE = \angle AEG, \therefore AE = AG.$$

$$\because EF = \sqrt{BE^2 + BF^2} = \sqrt{2^2 + 2^2} = 2\sqrt{2}, BF = GF = 2,$$

$$\therefore GE = EF - GF = 2\sqrt{2} - 2.$$

$$\because \triangle ABE \cong \triangle ADE, \therefore BE = DE = 2.$$

$$\because AG \parallel CD, \therefore \triangle AGE \sim \triangle CDE,$$

$$\therefore \frac{AG}{CD} = \frac{GE}{DE} = \frac{2\sqrt{2} - 2}{2} = \sqrt{2} - 1, \therefore \frac{AE}{AB} = \sqrt{2} - 1.$$

专题(8) 相似与函数

1. $(4, 3)$ 或 $(-8, -3)$

2. 解: 设 $E(t, \frac{1}{2}t + 2)$, 当 $y = 0$ 时, $\frac{1}{2}x + 2 = 0$,

$$\text{解得 } x = -4, \therefore A(-4, 0).$$

$$\text{当 } x = 0 \text{ 时, } y = \frac{1}{2}x + 2 = 2, \therefore B(0, 2).$$

$$\because C(4, 4), CD \perp x \text{ 轴},$$

$$\therefore CD = 4, BC = \sqrt{4^2 + (4-2)^2} = 2\sqrt{5}.$$

$$\because CD \parallel OB, \therefore \angle EBO = \angle BCD,$$

$$\therefore \text{当 } \frac{BE}{CB} = \frac{BO}{CD} \text{ 时, } \triangle BEO \sim \triangle CBD, \text{ 即 } \frac{BE}{2\sqrt{5}} = \frac{2}{4},$$

$$\text{解得 } BE = \sqrt{5}, \therefore t^2 + \left(\frac{1}{2}t + 2 - 2\right)^2 = 5,$$

$$\text{解得 } t_1 = 2 (\text{舍去}), t_2 = -2,$$

此时点 E 的坐标为 $(-2, 1)$;

$$\text{当 } \frac{BE}{CD} = \frac{BO}{CB} \text{ 时, } \triangle BEO \sim \triangle CDB,$$

$$\text{即 } \frac{BE}{4} = \frac{2}{2\sqrt{5}}, \text{ 解得 } BE = \frac{4\sqrt{5}}{5},$$

$$\therefore t^2 + \left(\frac{1}{2}t + 2 - 2\right)^2 = \frac{16}{5}, \text{ 解得 } t_1 = \frac{8}{5} (\text{舍去}), t_2 =$$

$$-\frac{8}{5}, \text{ 此时点 } E \text{ 的坐标为 } \left(-\frac{8}{5}, \frac{6}{5}\right).$$

综上所述, 点 E 的坐标为 $\left(-\frac{8}{5}, \frac{6}{5}\right)$ 或 $(-2, 1)$.

3. 解: (1) 将点 $A(2, 3)$ 代入 $y = \frac{k}{x}$ 中, 得 $k = 6$, $\therefore$ 反比例函

$$\text{数的表达式为 } y = \frac{6}{x}.$$

将点 $A(2, 3), B(0, 4)$ 代入 $y = mx + n$ 中,

$$\text{得 } \begin{cases} n = 4, \\ 2m + n = 3, \end{cases} \text{ 解得 } \begin{cases} m = -\frac{1}{2}, \\ n = 4, \end{cases}$$

$$\therefore \text{一次函数的表达式为 } y = -\frac{1}{2}x + 4.$$

图.

直平分 CF ,

方形,

$\therefore y$ 与 x 的函数表

).

,

B , 如答图,

$$= DH = \frac{\sqrt{2}}{2} AD =$$

$$\angle OCB = \angle OBC,$$

$$= 2\sqrt{10},$$

$$\therefore \angle DAC = \angle EAB, \frac{AD}{AC} = \frac{AE}{AB},$$

$$\therefore \triangle DAC \sim \triangle EAB, \therefore \frac{CD}{BE} = \frac{AC}{AB} = \frac{\sqrt{2}}{2}.$$

$$\textcircled{2} BE^2 + 2BC^2 = 2DE^2,$$

证明: $\because \triangle DAC \sim \triangle EAB$,

$$\therefore \angle EBA = \angle DCA = 90^\circ, \therefore BE^2 + AB^2 = AE^2.$$

$$\because AB^2 = 2BC^2, AE^2 = 2DE^2, \therefore BE^2 + 2BC^2 = 2DE^2.$$

(3) 解: 当点 D 在点 C 左侧时, $EC = \sqrt{10}$, 四边形 $ABED$

的面积为 $\frac{25}{2}$; 当点 D 在点 C 右侧时, $EC = \sqrt{34}$, 四边形

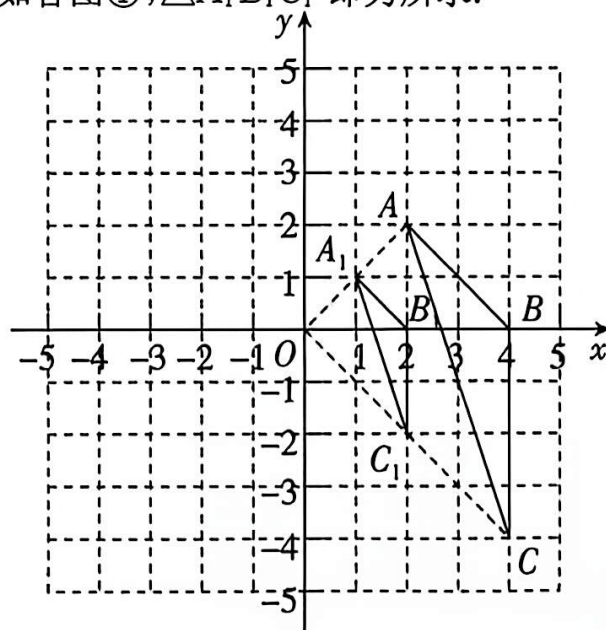
$ABDE$ 的面积为 $\frac{15}{2}$.

小结与思考

1. C 2. C 3. C 4. 18 5. $2\sqrt{5}-2$ 6. $\frac{5}{2}$ 7. ①③④

8. (1) 4

(2) 解: 如答图①, $\triangle A_1B_1C_1$ 即为所求.



第 8 题答图①

3亿人都在用的扫描App